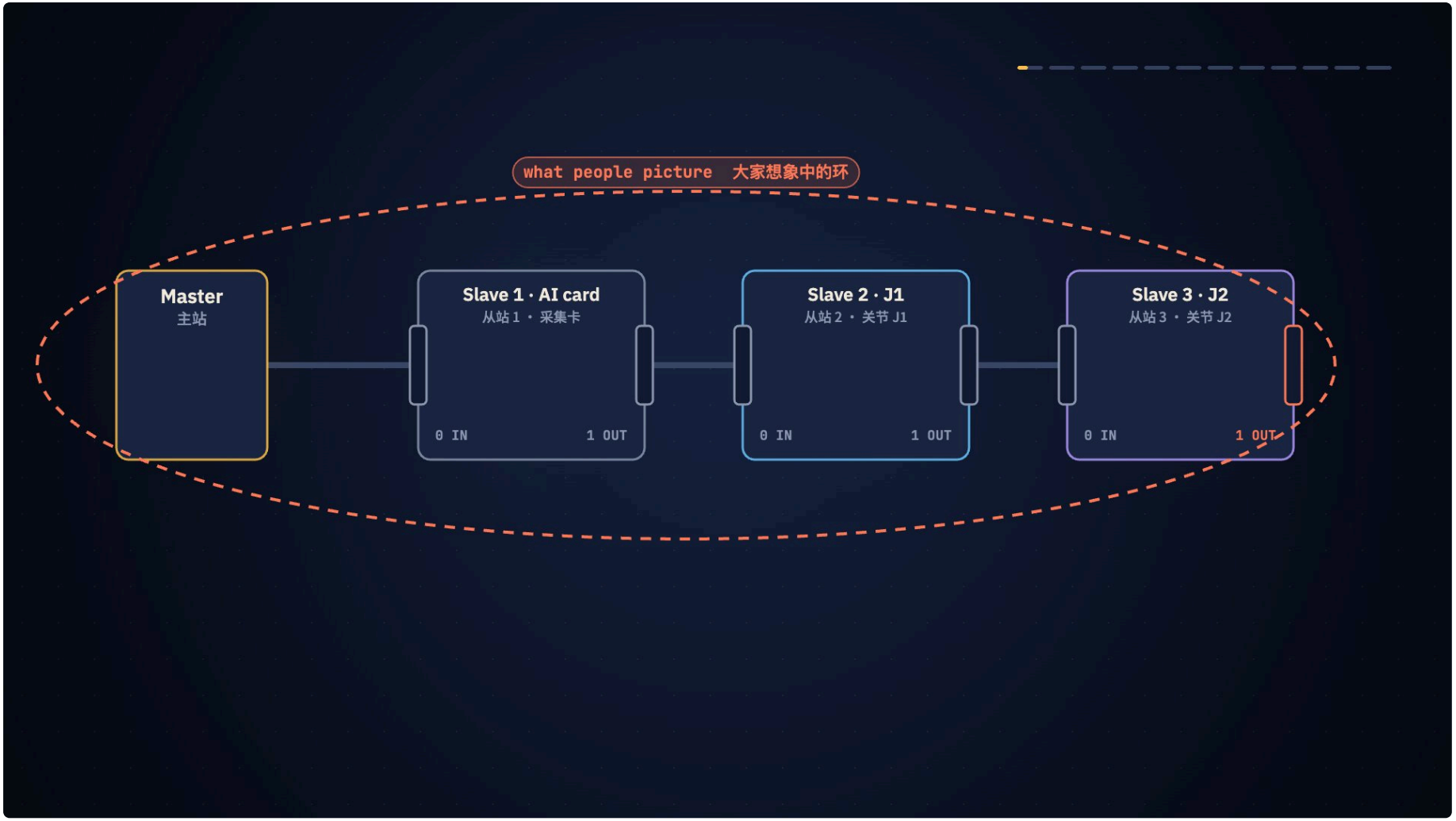


001 · 环在哪里? · Where is the ring? 0:00

上一集：同一帧先经过 J1，再经过 J2，相差 310 纳秒。

Last episode: one frame passed J1, then J2, 310 ns apart.

大家都说 EtherCAT 是一个环：帧绕一圈再回来。

People say EtherCAT is a ring: the frame goes round and comes back.



002 · 环在哪里? · Where is the ring? 0:13

可看看我们的总线：一根网线从主站出发，一个接一个，最后一个从站后面什么也没接。

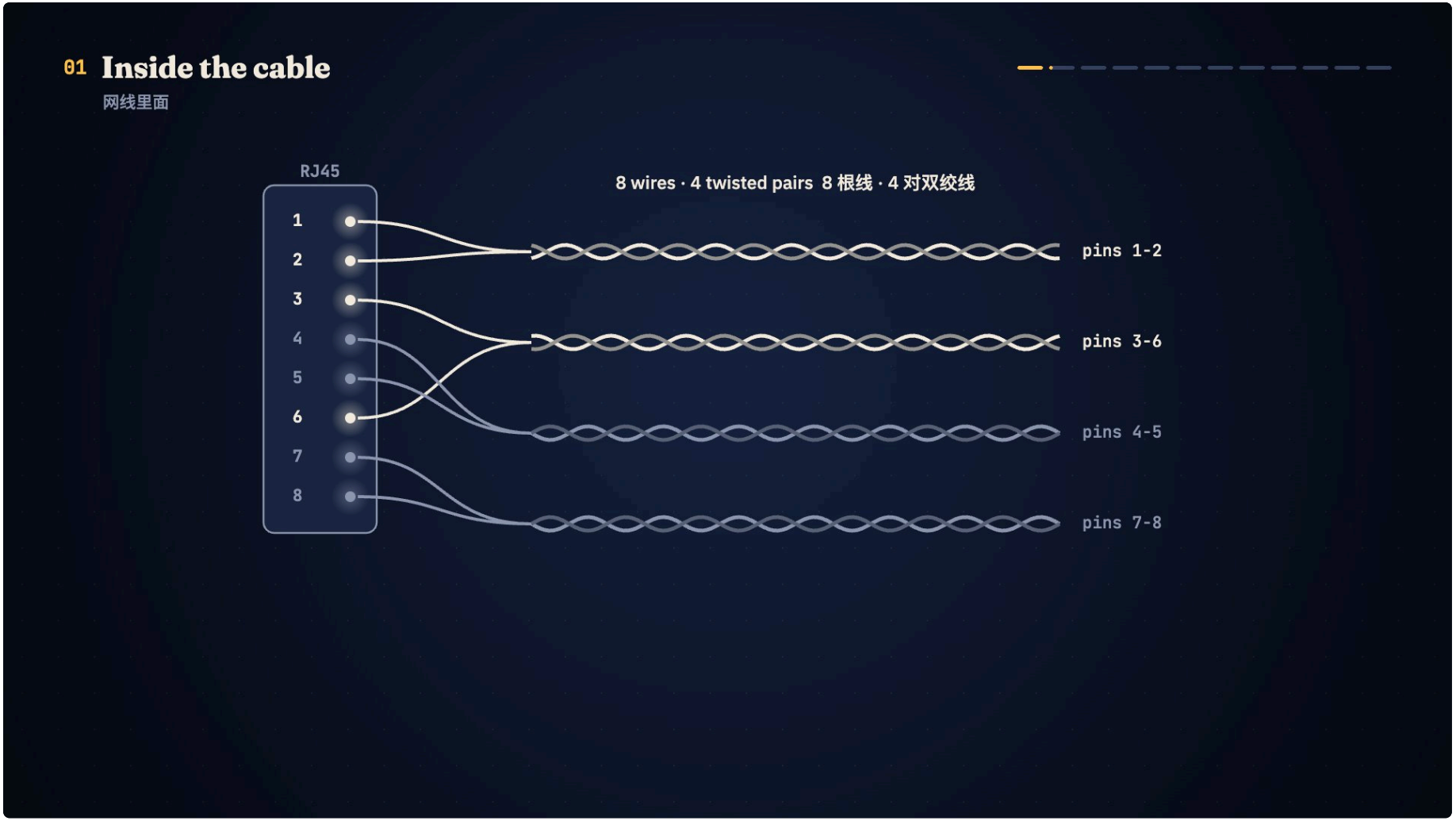
But look at our line. One cable from the master, slave to slave, and the last one goes nowhere.



003 · 环在哪里? · Where is the ring? 0:23

这是一条线，不是环。那环在哪里？

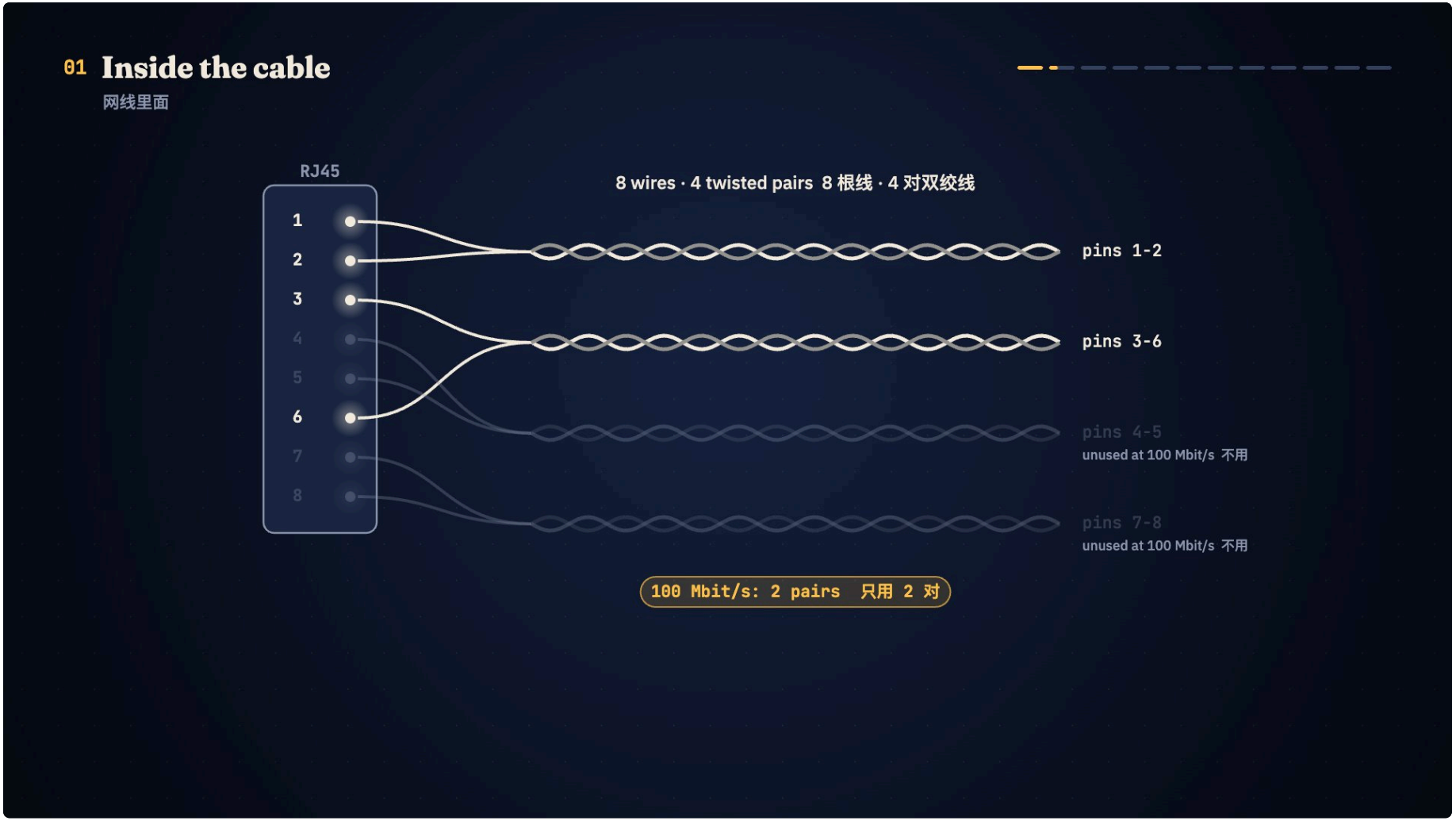
That is a line, not a ring. So where is the ring?



004 · 网线里面 · Inside the cable 0:31

剥开一根网线：8 根线，两两绞成 4 对。

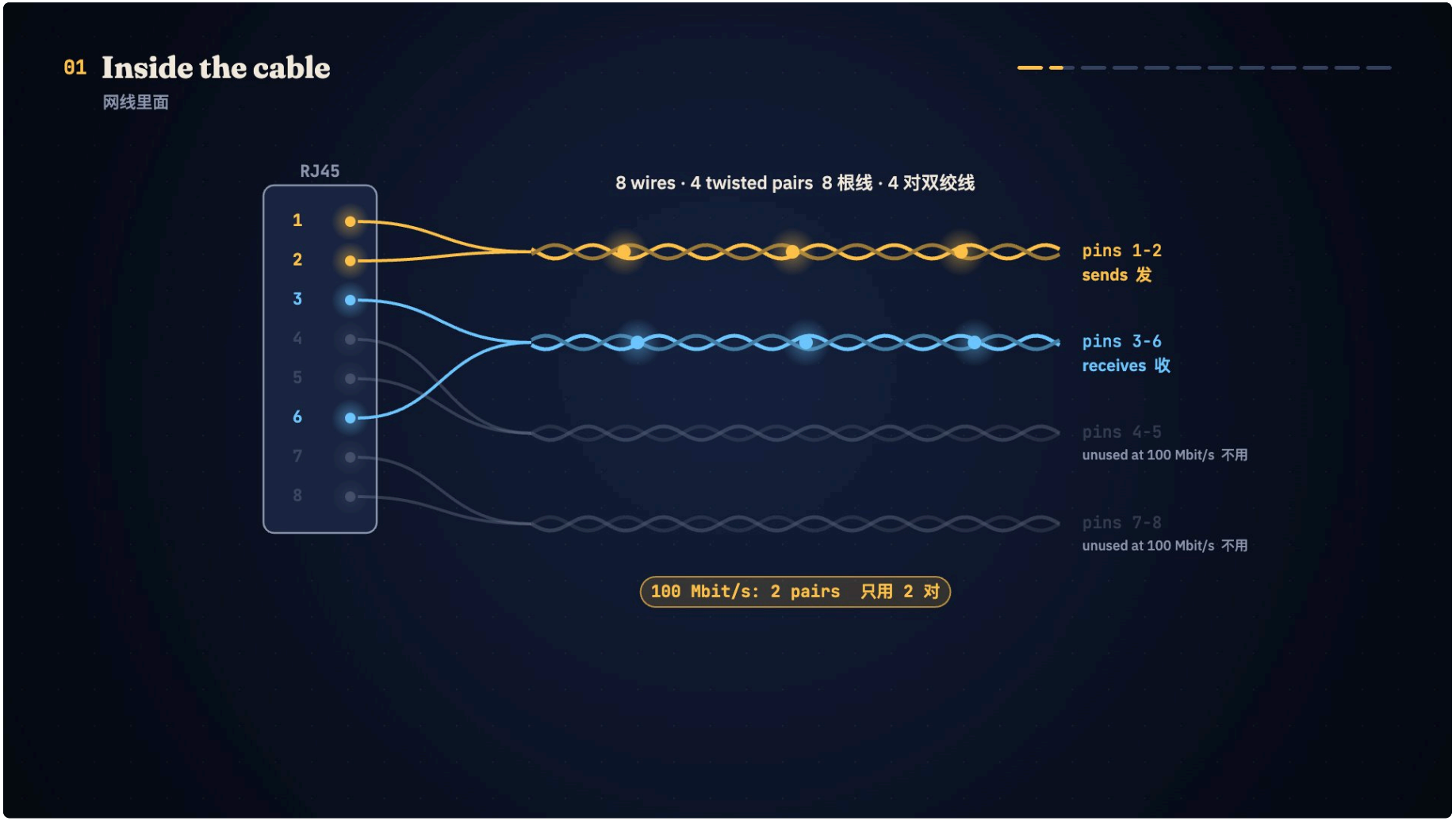
Open an Ethernet cable: eight wires, twisted into four pairs.



005 · 网线里面 · Inside the cable 0:37

EtherCAT 跑 100 Mbit/s，这个速率只用其中 2 对。

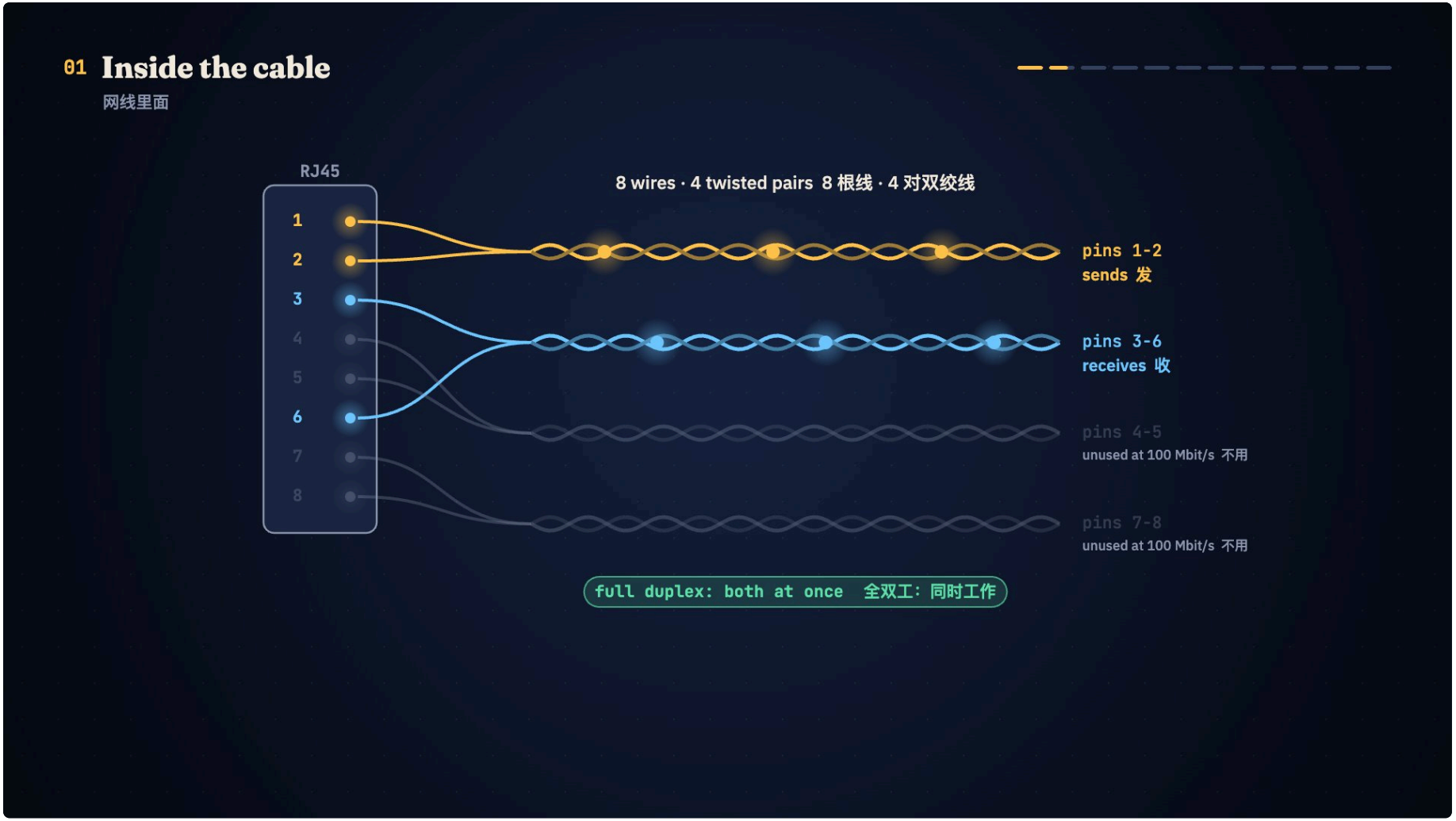
EtherCAT runs at 100 Mbit/s, and that speed uses only two of the pairs.



006 · 网线里面 · Inside the cable 0:44

1、2 脚是一对，负责发；3、6 脚是一对，负责收。

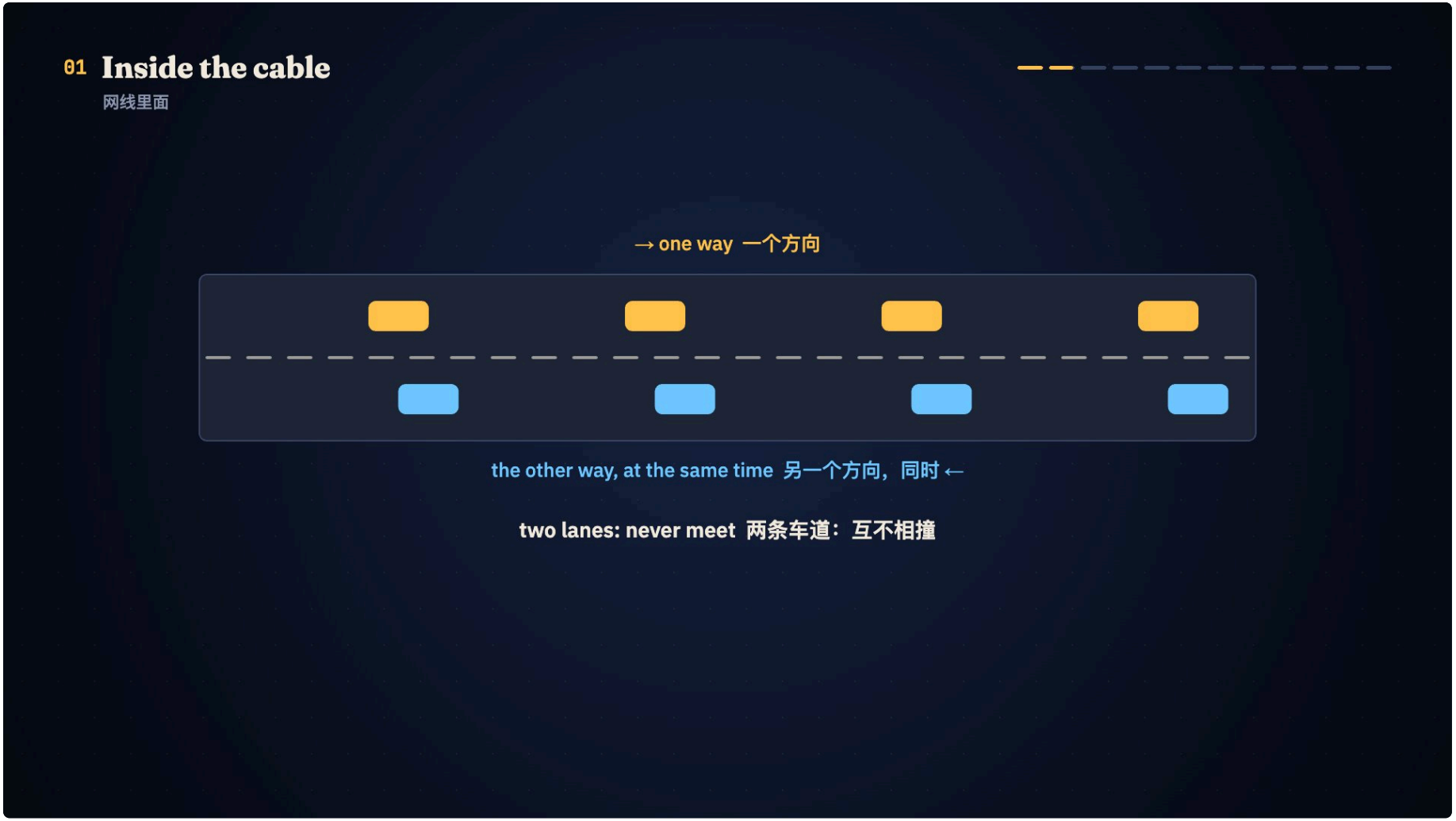
Pins 1 and 2: one pair that sends. Pins 3 and 6: one pair that receives.



007 · 网线里面 · Inside the cable 0:52

两对同时工作，这叫全双工。

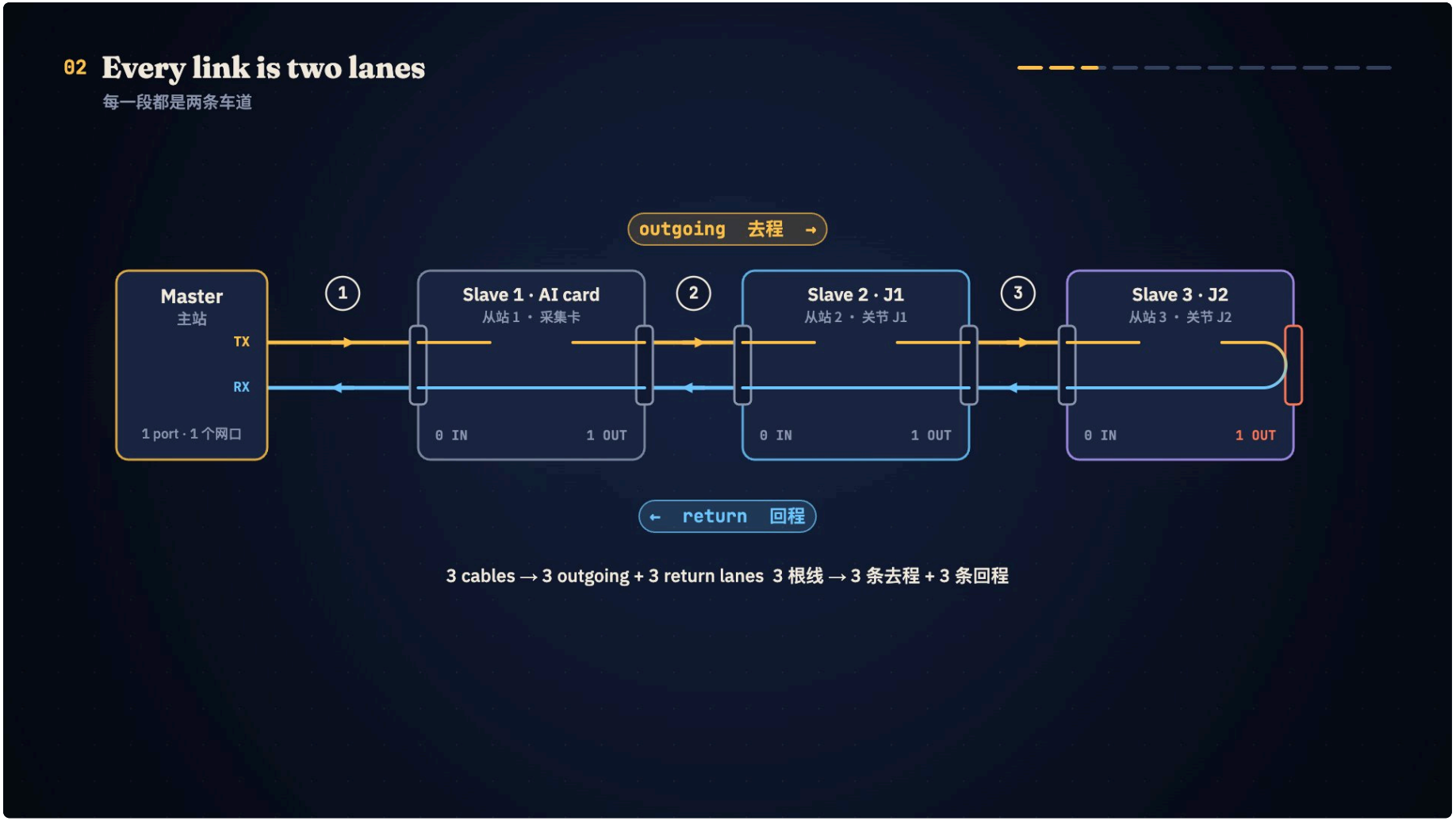
Both pairs work at the same time. This is called full duplex.



008 · 网线里面 · Inside the cable 0:58

就像一条双车道的路：两个方向同时通车，互不相撞。

Like a road with two lanes: traffic goes both ways at once, and never meets.



009 · 每一段都是两条车道 · Every link is two lanes 1:08

按这个样子重画我们的总线：每根网线都变成两条车道。

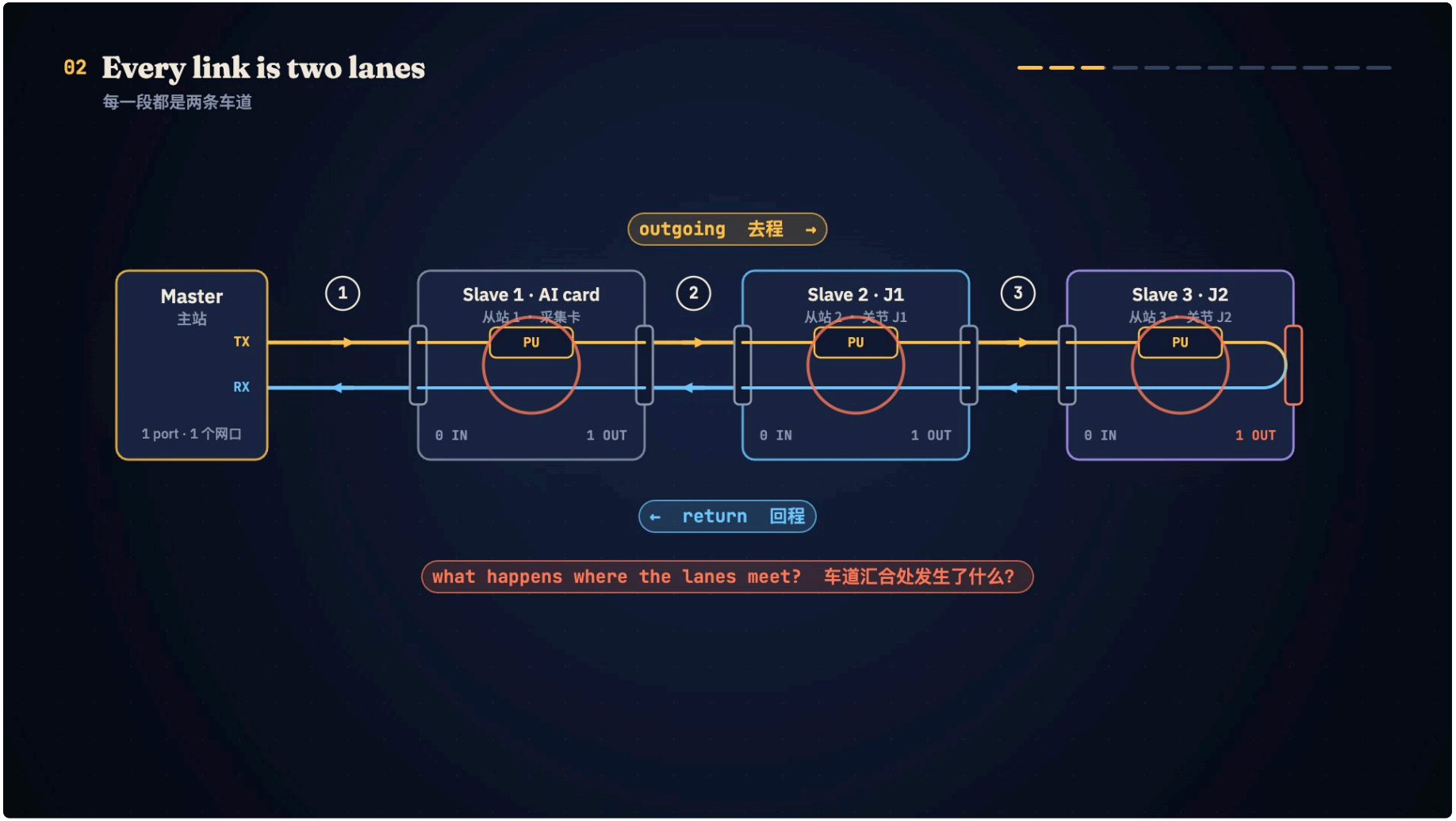
Now draw our line that way: every cable becomes two lanes.

离开主站的那条叫去程，回主站的那条叫回程。

The lane away from the master we call outgoing. The lane back we call return.

三根网线，就有三条去程、三条回程。

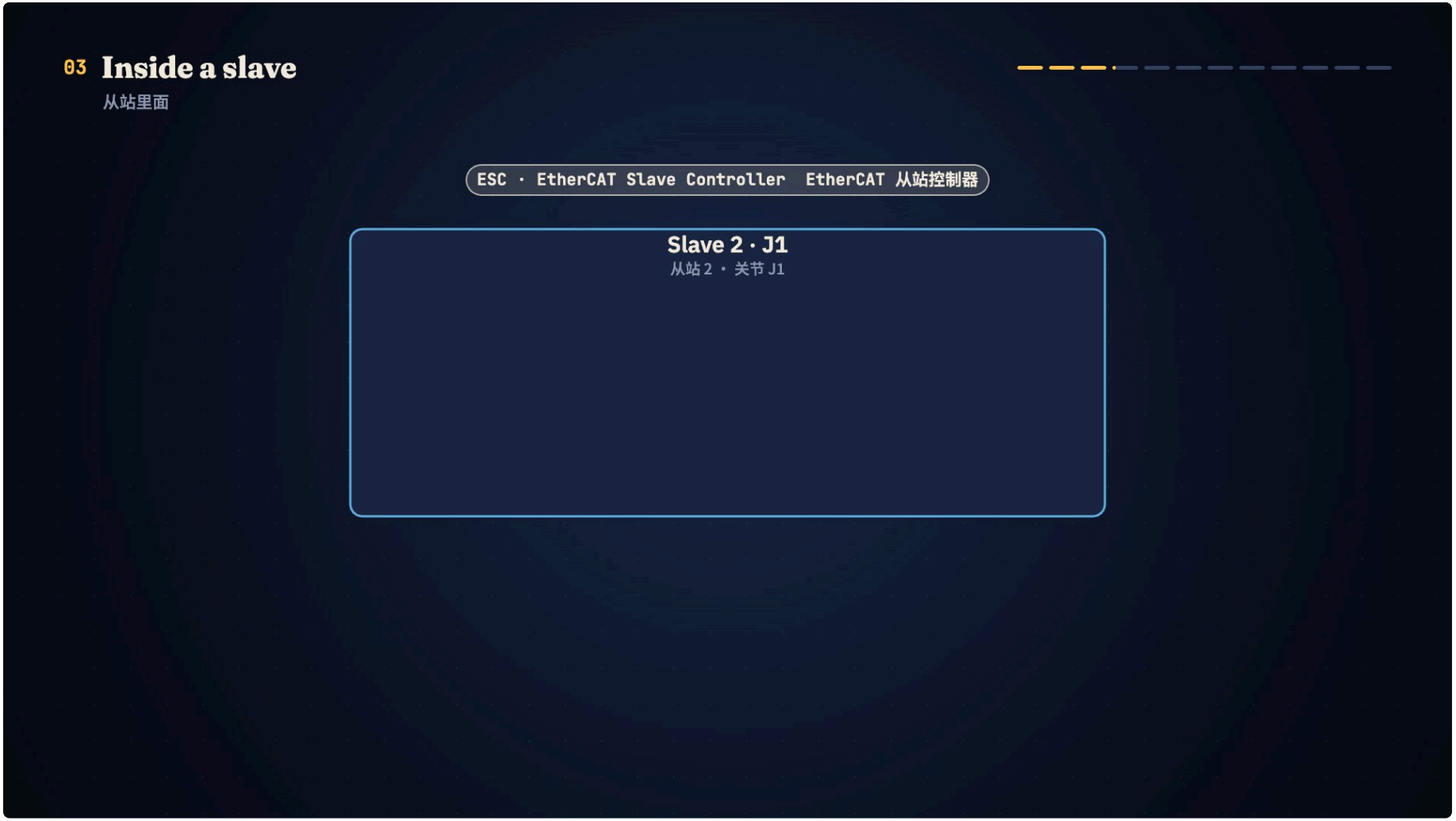
Three cables, so three outgoing lanes and three return lanes.



010 · 每一段都是两条车道 · Every link is two lanes 1:28

问题变成了：车道在每个从站里汇合，那里发生了什么？

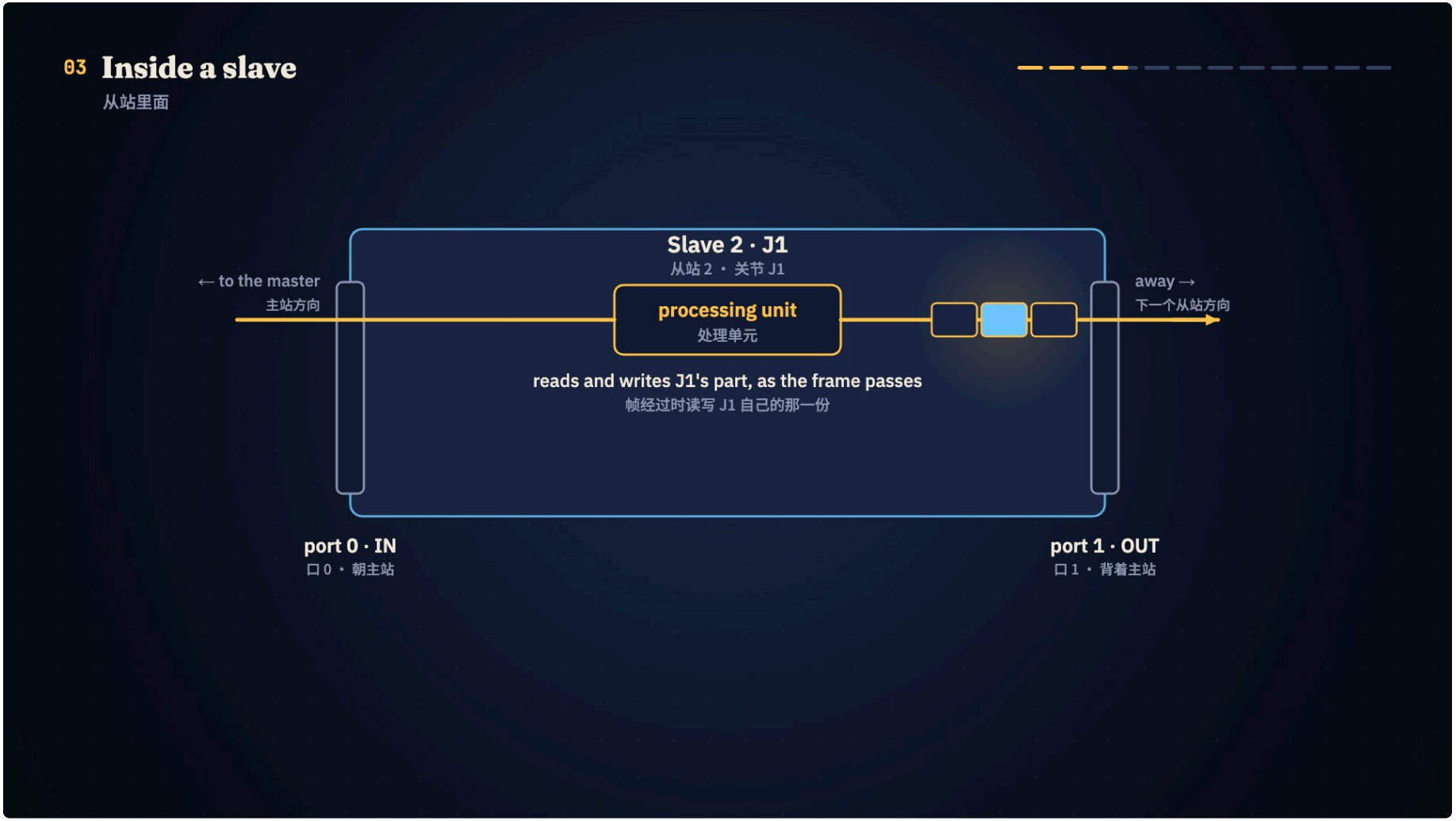
The question is now: what happens inside each slave, where the lanes meet?



011 · 从站里面 · Inside a slave 1:37

每个从站里有一颗专管 EtherCAT 的芯片，叫 ESC（EtherCAT 从站控制器）。

Every slave has a chip for EtherCAT, the ESC: EtherCAT Slave Controller.



012 · 从站里面 · Inside a slave 1:43

我们的 ESC 有两个口。口 0 朝着主站，标 IN；口 1 背着主站，标 OUT。

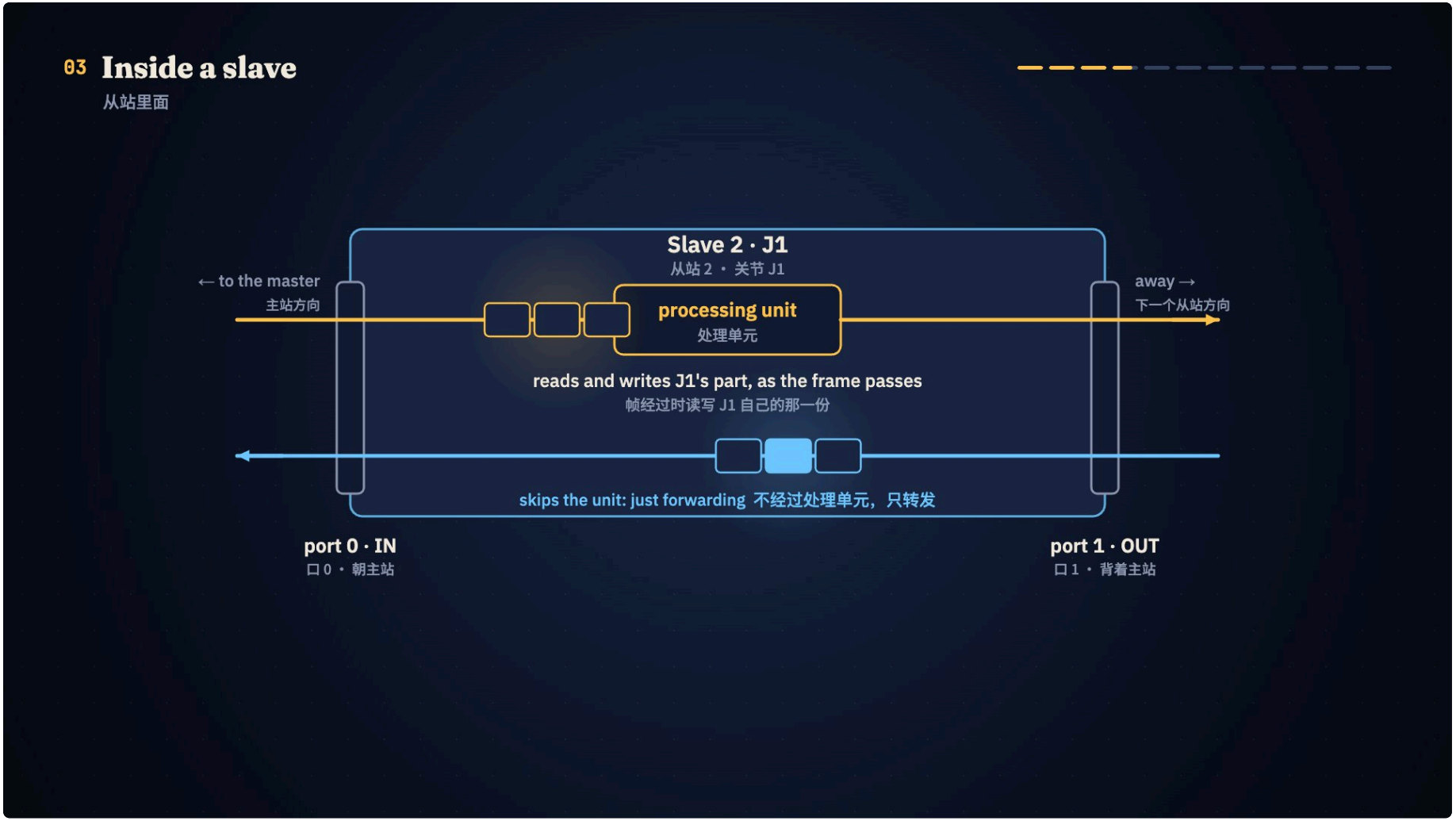
Our ESCs have two ports. Port 0 faces the master: IN. Port 1 faces away: OUT.

从口 0 进来的帧，先经过处理单元，再从口 1 出去。

A frame arriving at port 0 goes through the processing unit, then out of port 1.

处理单元就在帧经过时，读写帧里属于这个从站的数据。

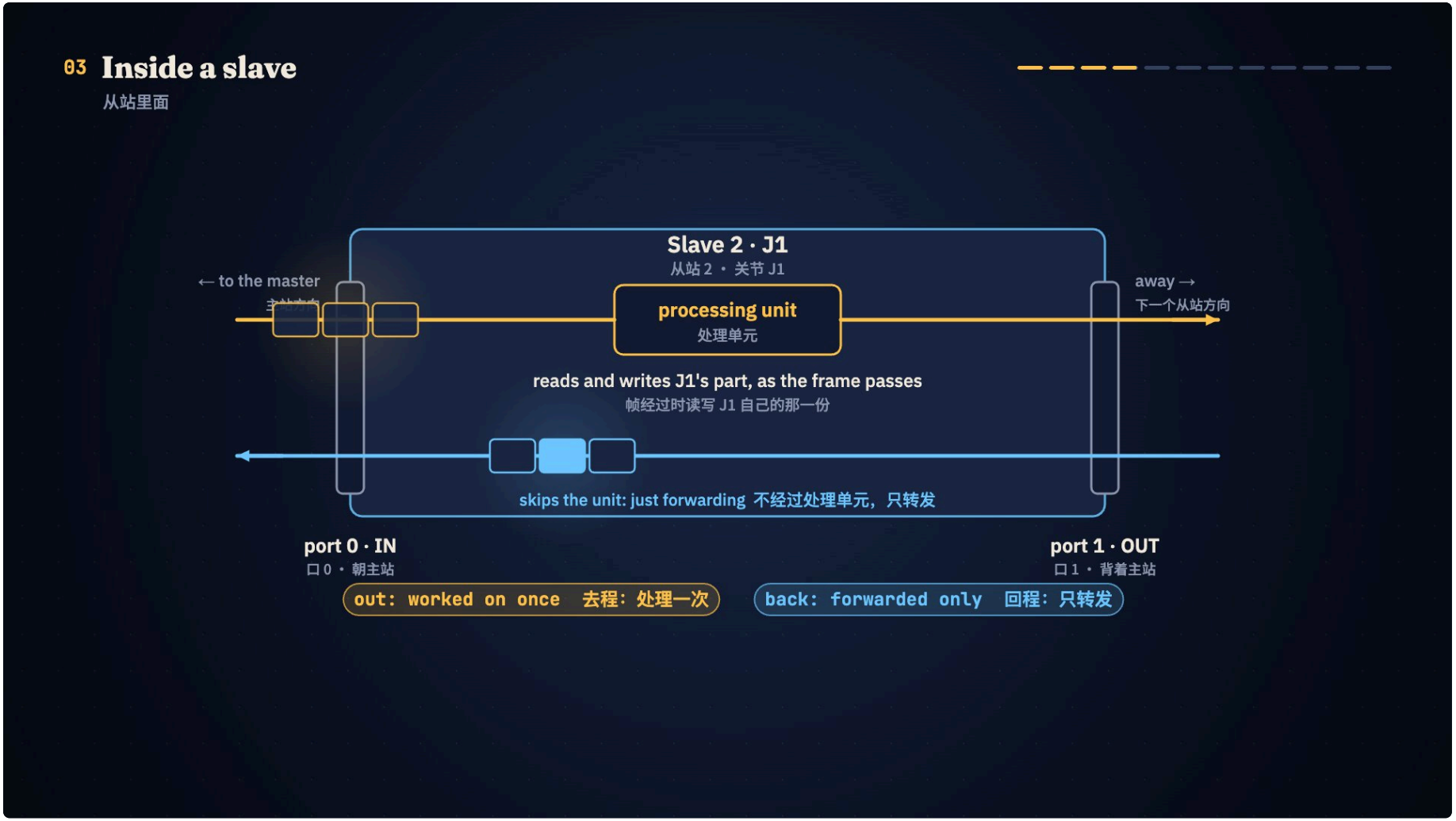
The processing unit reads and writes this slave's data in the frame, as it passes.



013 · 从站里面 · Inside a slave 2:07

从口 1 进来的帧不经过它，直接从口 0 出去。

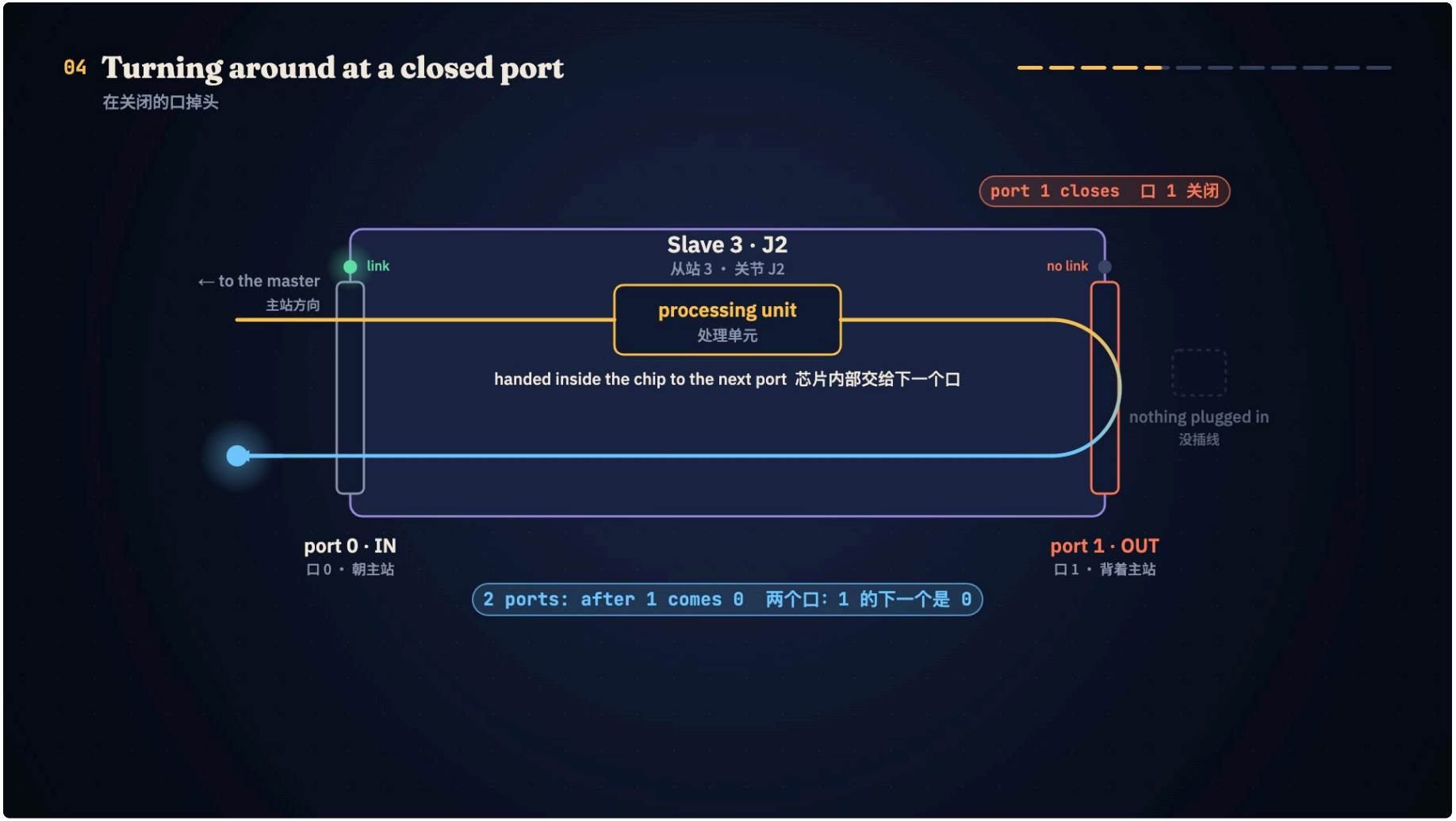
A frame arriving at port 1 skips it, and goes straight out of port 0.



014 · 从站里面 · Inside a slave 2:15

所以每个从站只在去程上处理一次这一帧，回程只是转发。

So each slave works on the frame once, on the way out. The return trip is just forwarding.



015 · 在关闭的口掉头 · Turning around at a closed port 2:26

再看最后一个从站 J2：它的口 1 上什么也没插。

Now the last slave, J2. Nothing is plugged into its port 1.

没插线就没有链路。ESC 发现了，就把这个口关掉。

No cable means no link. The ESC sees that and closes the port.

关闭的口不往外发，帧在芯片内部直接交给下一个口。

A closed port does not send. The frame is passed inside the chip to the next port instead.

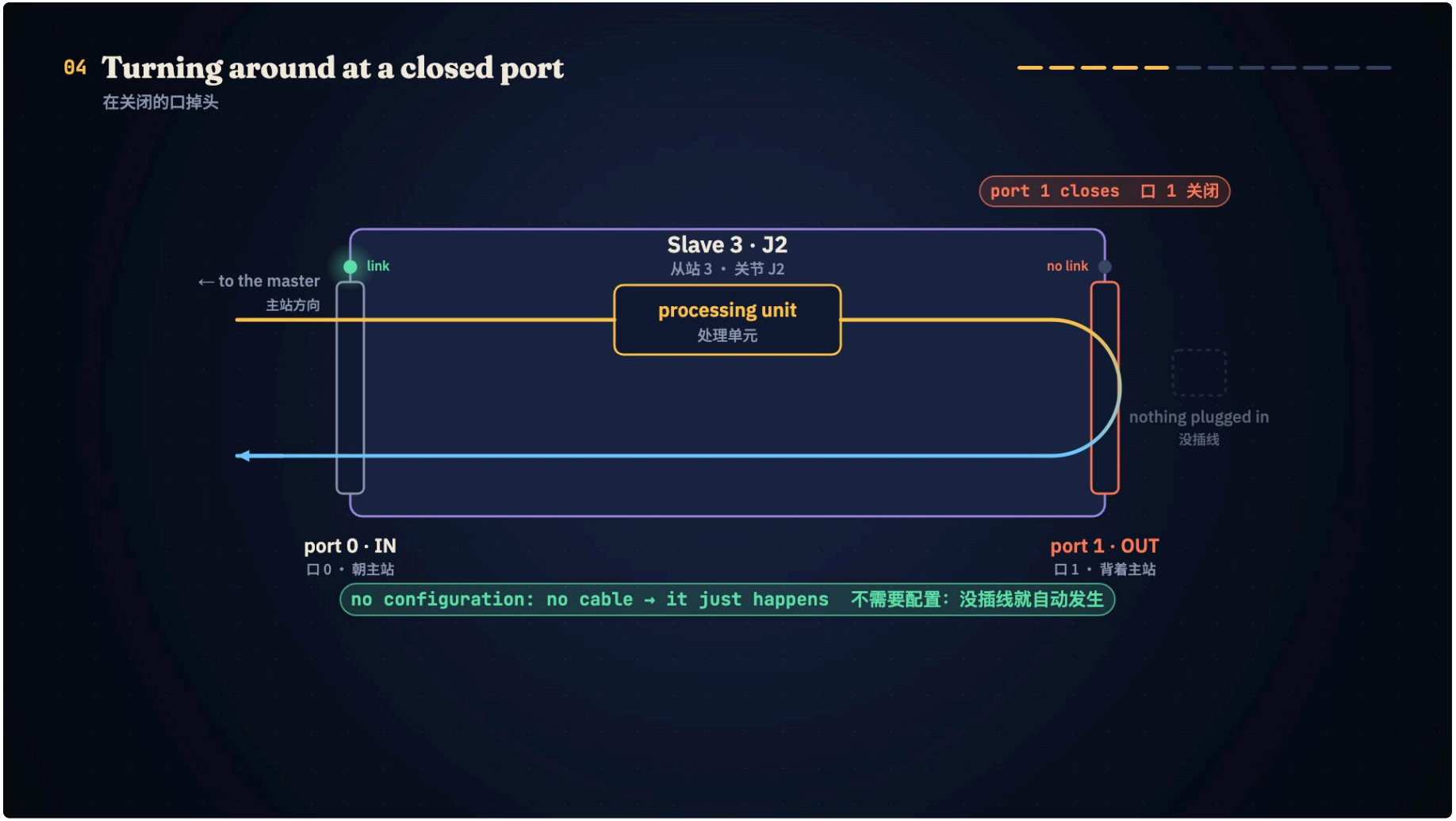
两个口的芯片，口 1 的下一个就是口 0，于是帧掉头往回走。



016 · 在关闭的口掉头 · Turning around at a closed port 2:58

就像一条死胡同，尽头有个掉头的圆环。

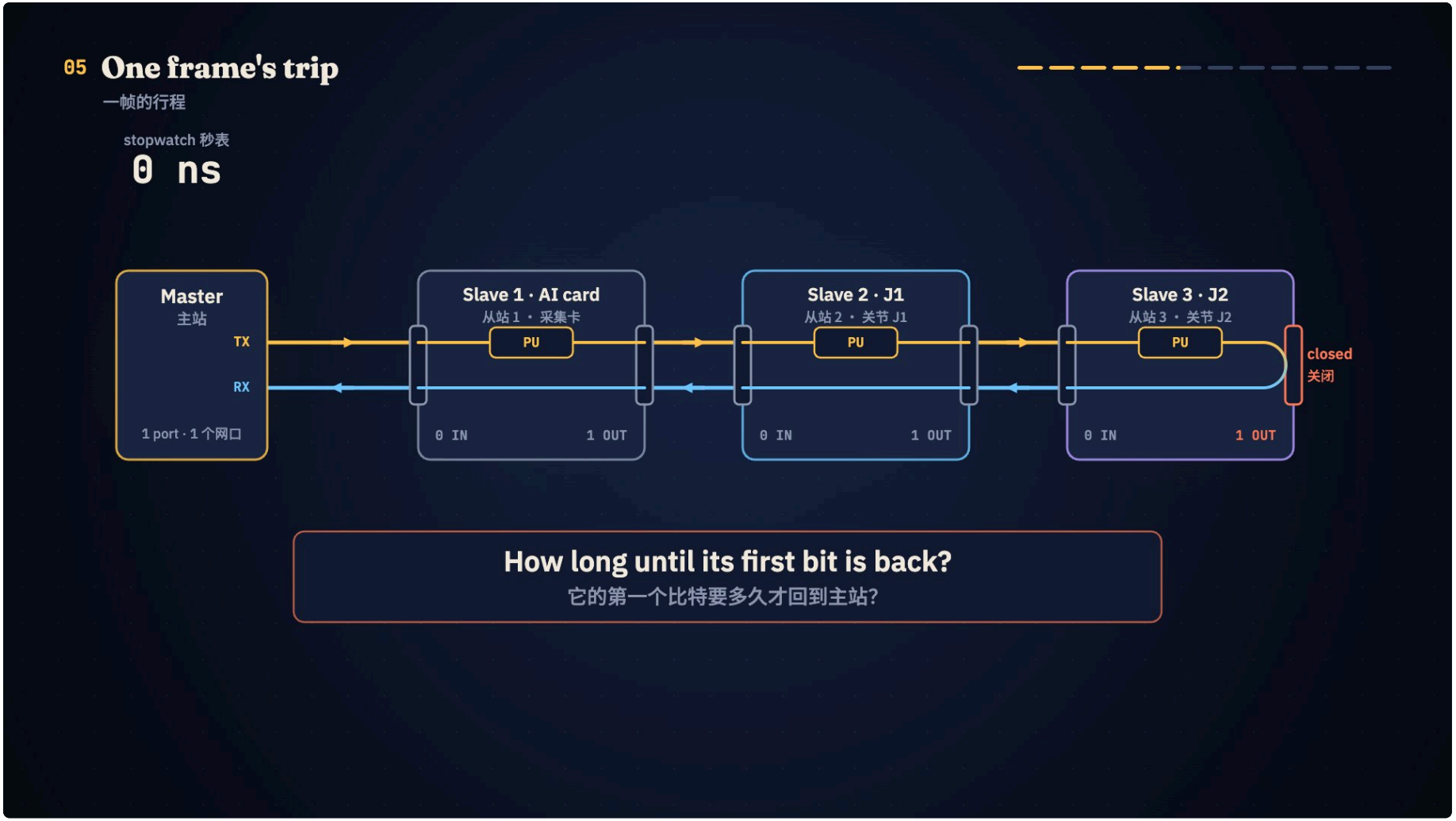
Like a dead-end street with a turning circle at the end.



017 · 在关闭的口掉头 · Turning around at a closed port 3:04

这不需要谁去配置：没插线，它就自动发生。

Nobody configured this. It happens because the cable is missing.



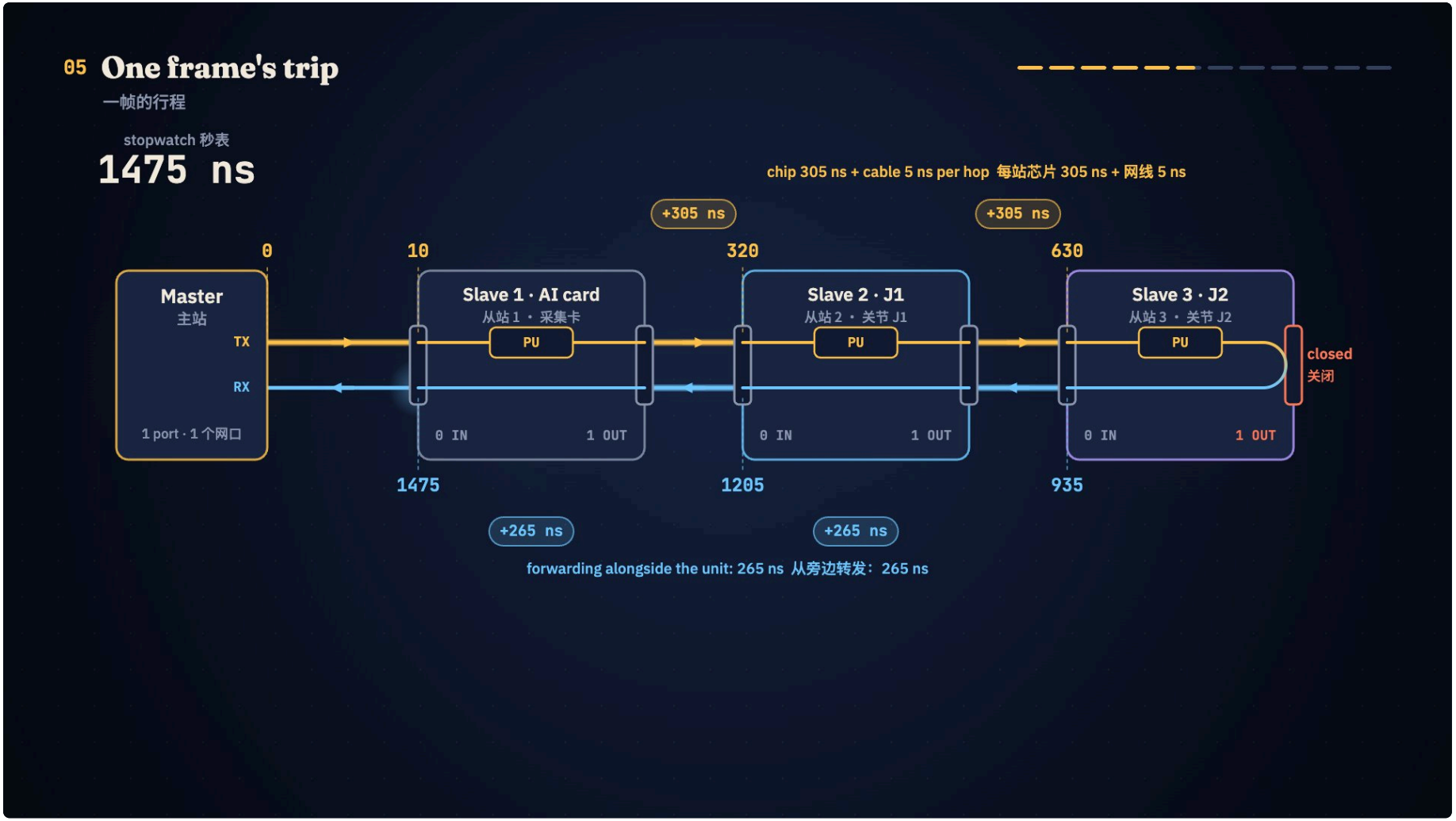
018 · 一帧的行程 · One frame's trip 3:11

现在拿着秒表，跟着一帧从主站出发。

Now follow one frame from the master, with a stopwatch.

先猜：它的第一个比特要多久才回到主站?

Guess first: how long until its first bit is back at the master?



019 · 一帧的行程 · One frame's trip 3:26

10 纳秒：到达从站 1。2 米网线，模拟的每米 5 纳秒。

10 ns: it reaches slave 1. 2 m of cable at a simulated 5 ns per metre.

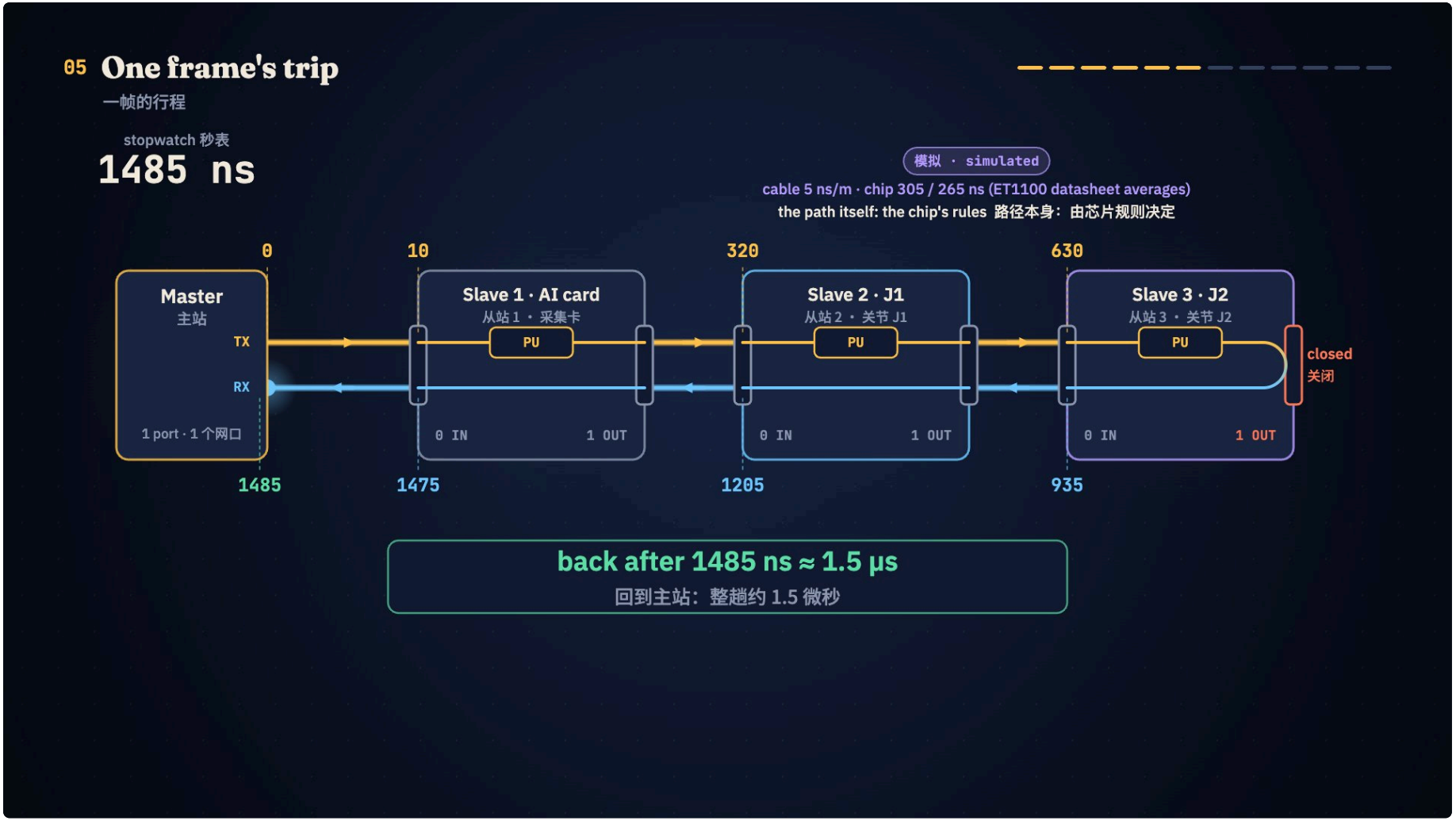
320 纳秒到 J1，630 纳秒到 J2。

320 ns: it reaches J1. 630 ns: it reaches J2.

去程上每个从站添 305 纳秒：ET1100 数据手册里经过处理单元的平均值。

Each slave on the way out adds 305 ns, the ET1100 datasheet's average through the processing unit.

935 纳秒：掉过头，又从 J2 出来



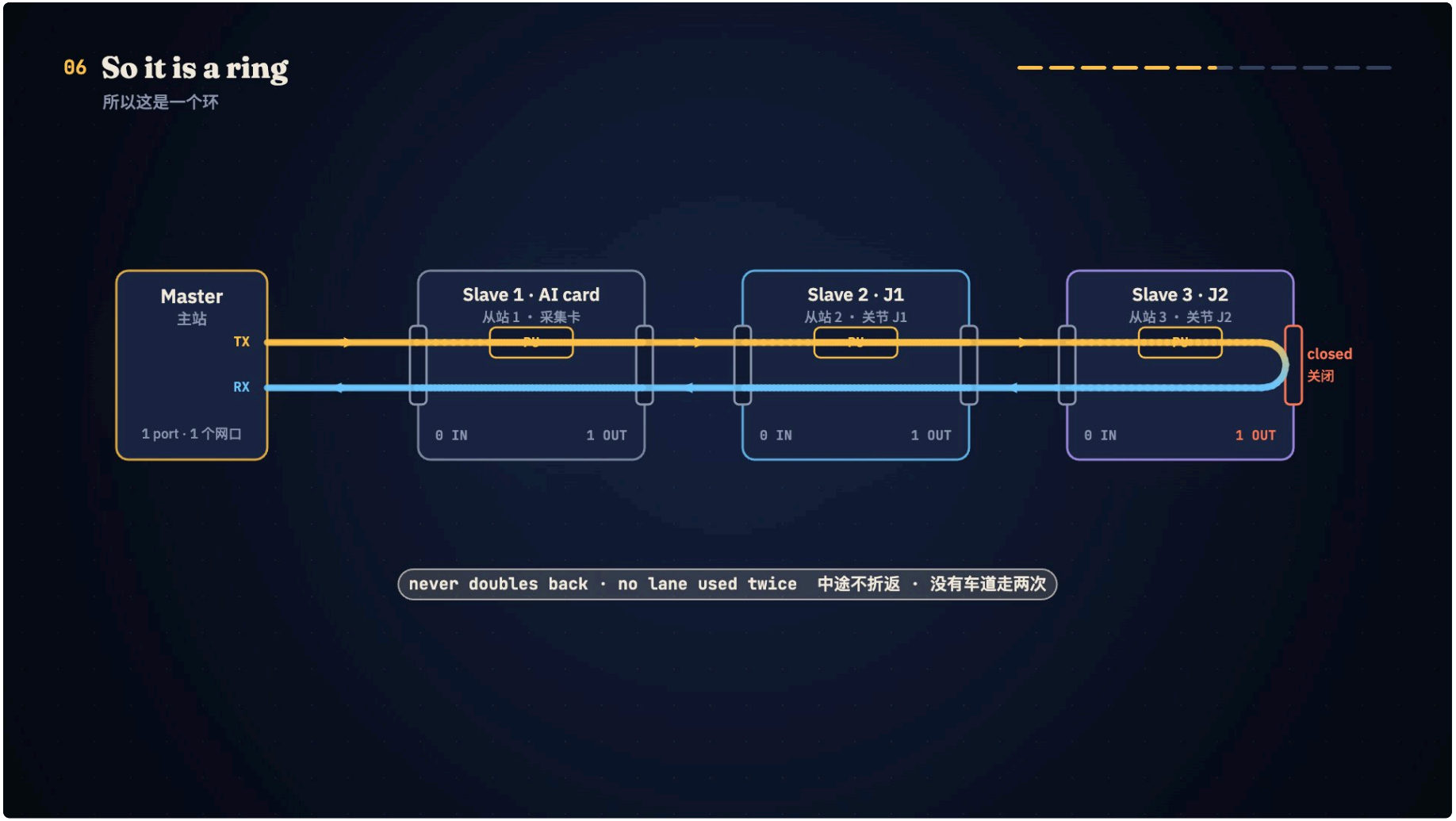
020 · 一帧的行程 · One frame's trip 4:07

1485 纳秒：回到主站。整趟约 1.5 微秒。

1485 ns: back at the master. About 1.5 µs for the whole trip.

网线和芯片延迟是模拟的；帧走哪条路，是芯片规则决定的。

The cable and chip delays are simulated. The path the frame takes is what the chip's rules say.



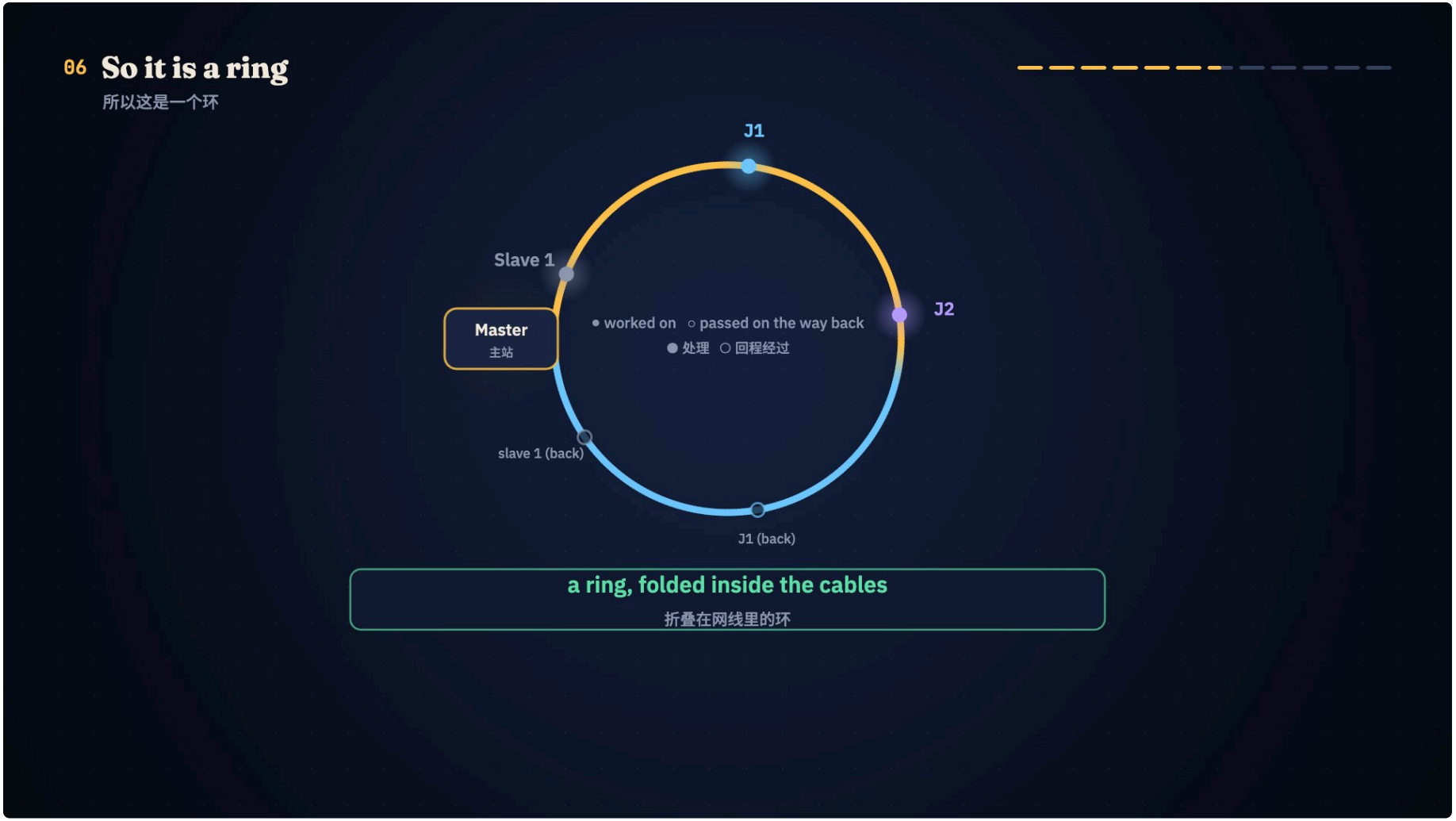
021 · 所以这是一个环 · So it is a ring 4:25

现在看这条路径：去程走遍每一条去程车道，回程走遍每一条回程车道。

Now look at the path: out on every outgoing lane, back on every return lane.

中途不折返，也没有哪条车道走两次。

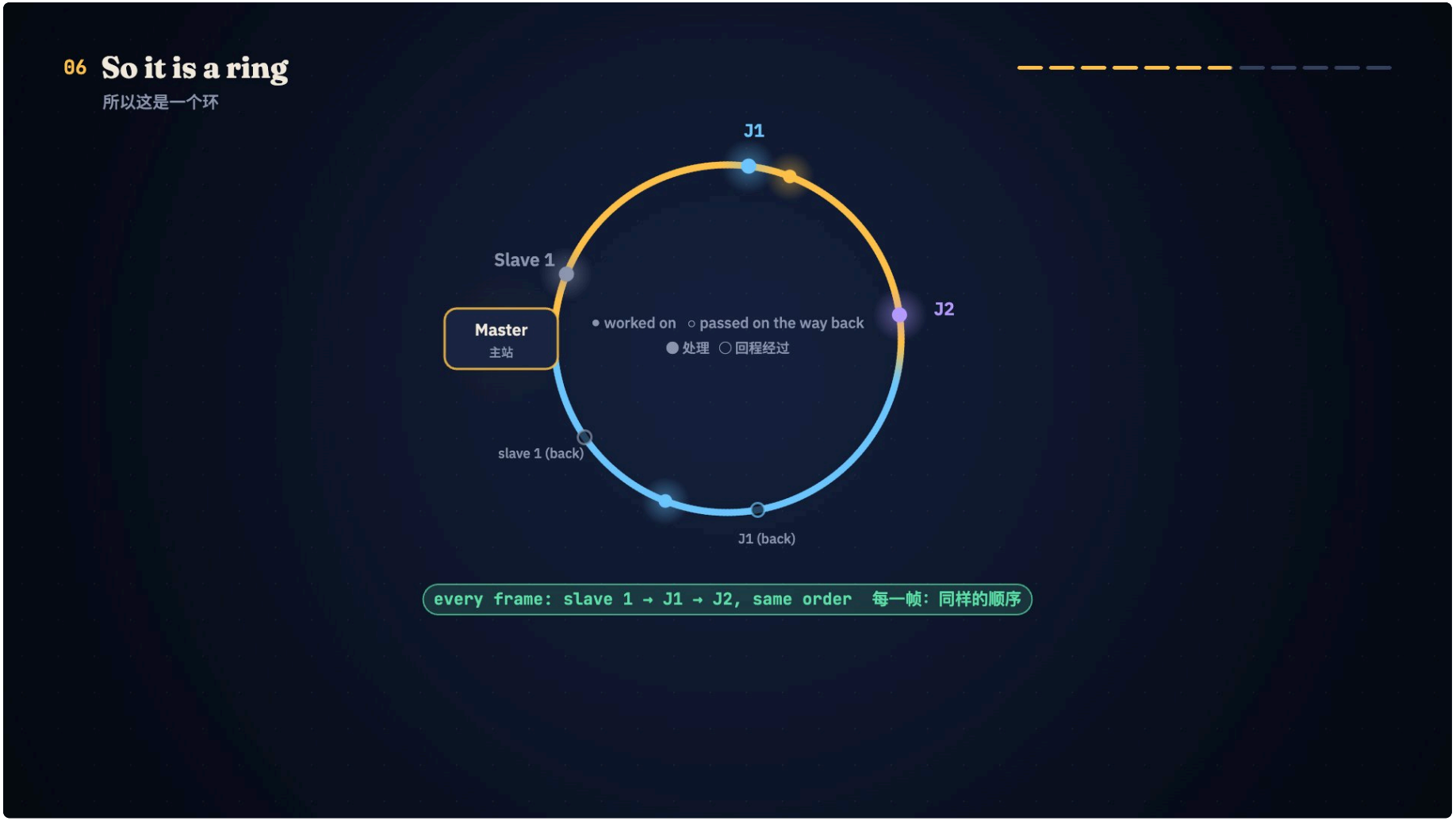
It never doubles back halfway, and no lane is used twice.



022 · 所以这是一个环 · So it is a ring 4:39

这就是一个环，只不过折叠在网线里面。

That is a ring. It is just folded inside the cables.



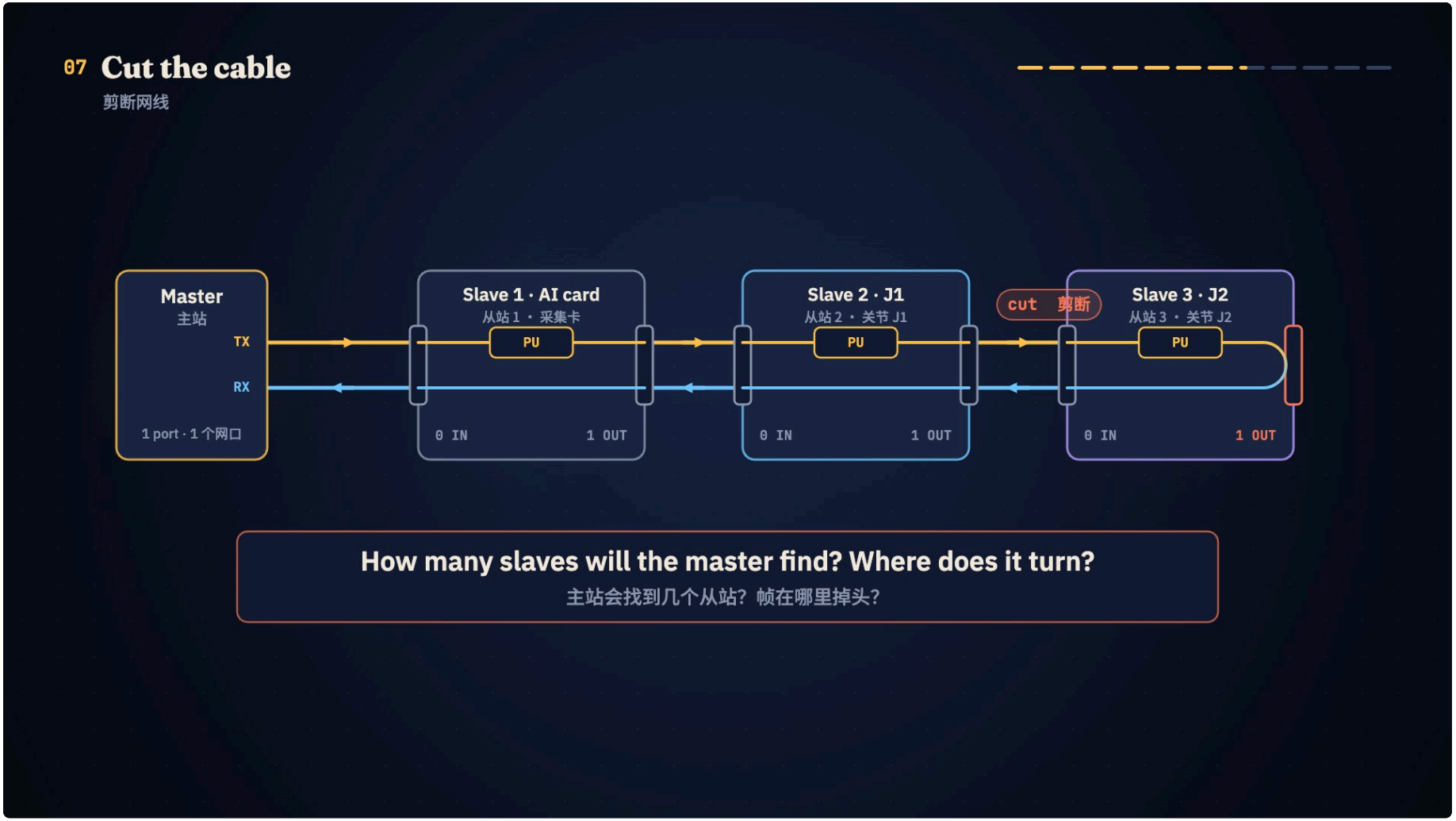
023 · 所以这是一个环 · So it is a ring 4:45

主站只需要一个网口：一对线发，另一对线收。

The master needs one Ethernet port: it sends on one pair and receives on the other.

而且每一帧都按同样的顺序经过每一个从站，每一次都一样。

And every frame visits every slave, in the same order, every time.



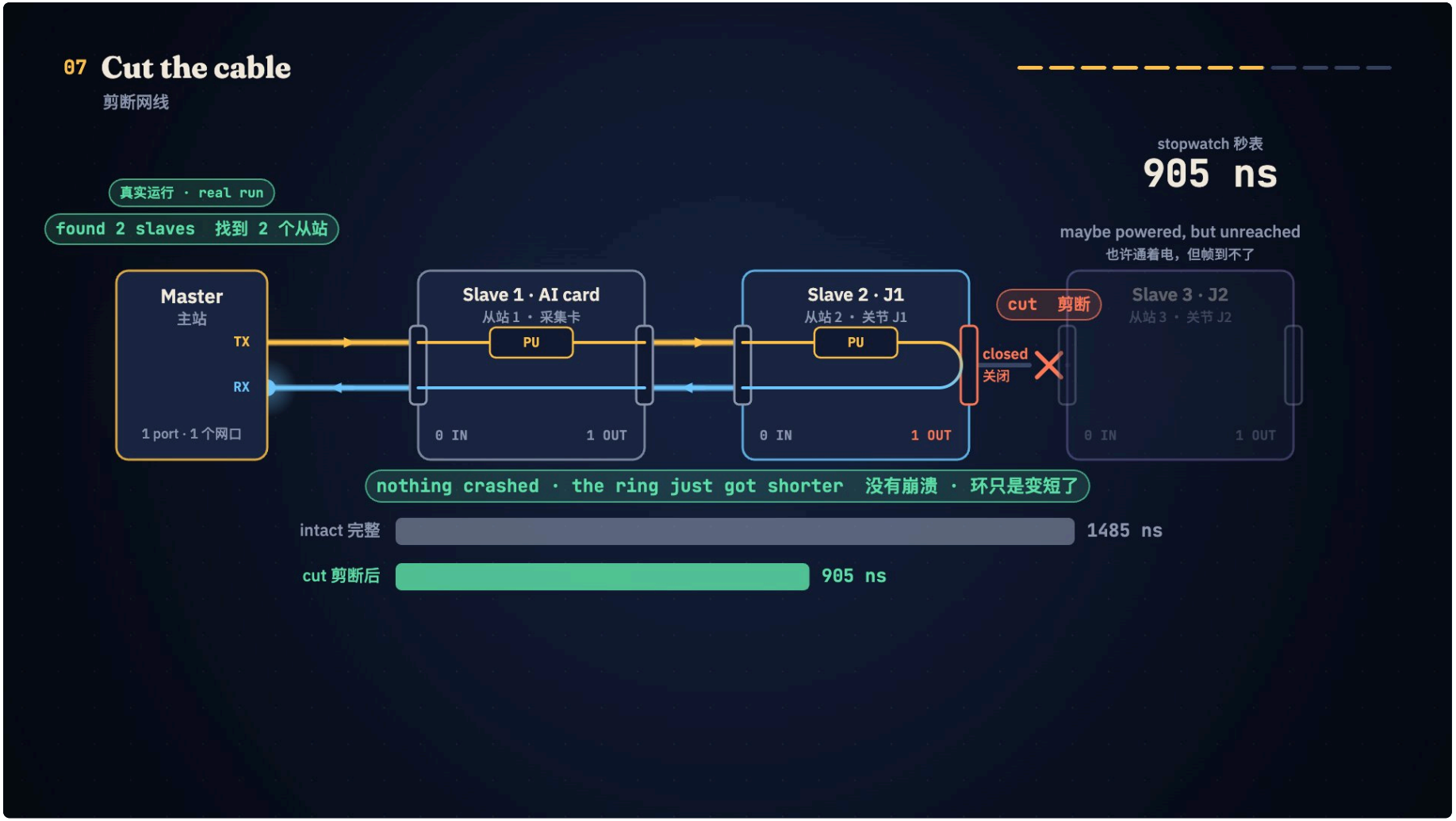
024 · 剪断网线 · Cut the cable 5:03

现在把 J1 和 J2 之间的网线剪断。

Now cut the cable between J1 and J2.

猜一猜：主站会找到几个从站？帧在哪里掉头？

Guess: how many slaves will the master find? And where does the frame turn around?



025 · 剪断网线 · Cut the cable 5:17

J1 的口 1 失去链路，于是关闭。帧现在在 J1 掉头。

J1's port 1 loses its link, so it closes. The frame now turns at J1.

主站找到 2 个从站。J2 也许还通着电，但没有帧能到它那里。

The master finds 2 slaves. J2 may still be powered, but no frame reaches it.

行程变短：905 纳秒，而不是 1485 纳秒。

The trip is shorter: 905 ns instead of 1485 ns.

没有崩溃，也不用重新配置：环只是变短了



026 · SOEM 怎么看到的 · How SOEM sees it 5:46

主站怎么知道的？SOEM 里有两小段代码。

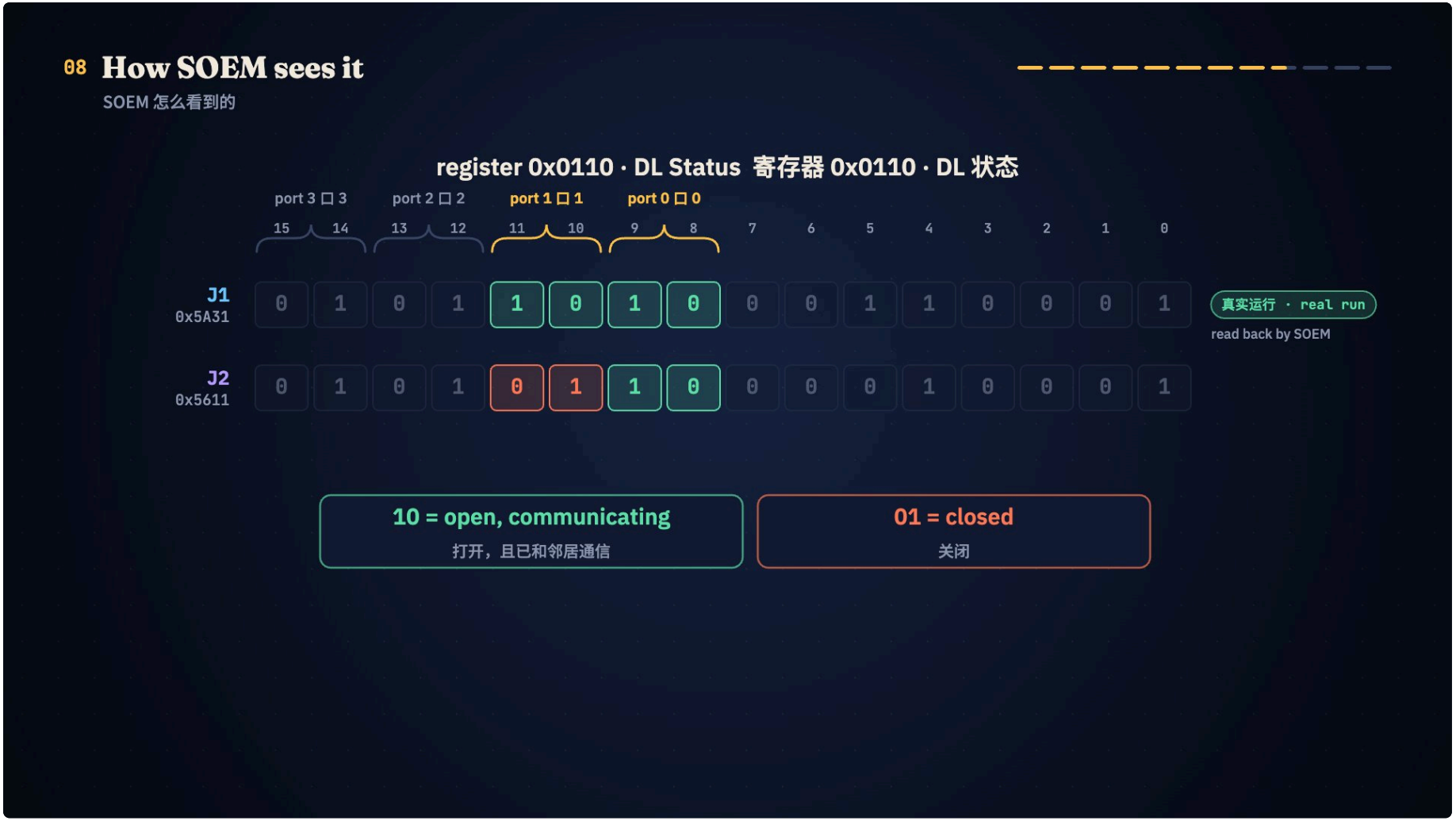
How does the master know? Two small pieces of SOEM.

先数人数：发一个广播读，每经过一个从站，帧里的计数器就加 1。

First it counts: one broadcast read, and every slave it passes adds 1 to a counter in the frame.

计数器回来是 3，剪断后是 2。这个计数器第 3 集还会细讲。

The counter comes back as 3, or as 2 after the cut. Episode 3 says more about this counter.



027 · SOEM 怎么看到的 · How SOEM sees it 6:10

然后逐个读从站的寄存器 0x0110，DL 状态，也就是各个口的状态。

Then it asks each slave for register 0x0110, DL Status: the state of its ports.

每个口两位。10 这个组合的意思是：打开了，而且和邻居建立了通信。

Two bits per port. Pattern 10 means: open, and communication established with the neighbour.

08 **How SOEM sees it**

SOEM 怎么看到的

soem/src/ec_config.c:279-292 ecx_config_init

```
topology = ecx_FPRDw(&context->port, configadr, ECT_REG_DLSTAT, EC_TIMEOUTRET3); /* extract topology from DL status */
topology = etohs(topology);
h = 0;
b = 0;
if ((topology & 0x0300) == 0x0200) /* port0 open and communication established */
{
   h++;
   b |= 0x01;
}
if ((topology & 0x0c00) == 0x0800) /* port1 open and communication established */
{
   h++;
   b |= 0x02;
}
```

soem/src/ec_config.c:307

```
context->slavelist[slave].activeports = b;
```

028 · SOEM 怎么看到的 · How SOEM sees it 6:25

SOEM 挨个检查每个口，把打开的口记下来。

SOEM tests each port in turn, and remembers the open ones.



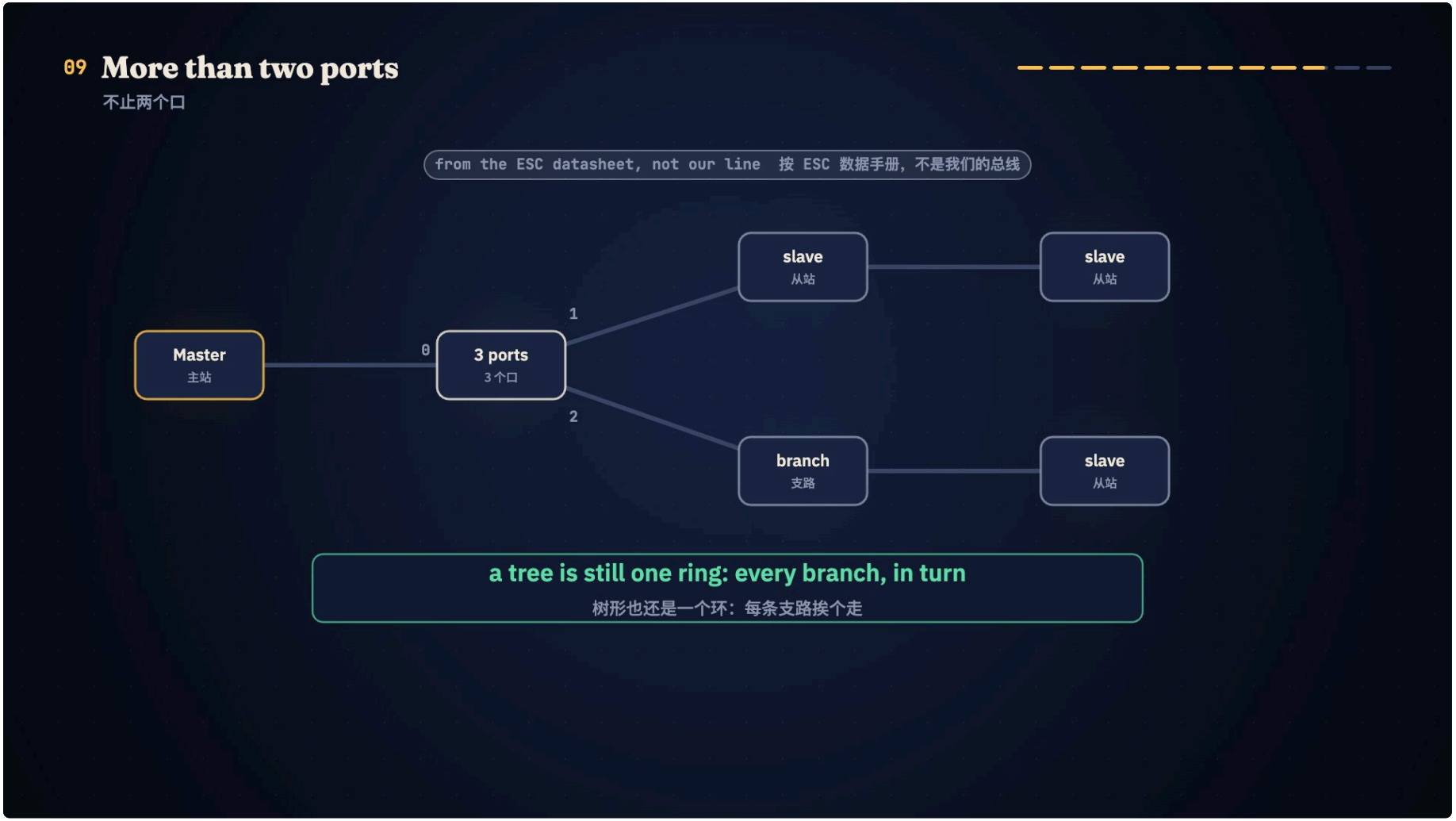
029 · SOEM 怎么看到的 · How SOEM sees it 6:31

完整的总线：从站 1 和 J1 两个口都开，J2 只开口 0。

Intact line: slave 1 and J1 show both ports open, J2 only port 0.

剪断后，J1 也只剩口 0 开着。哪些口开着，就告诉了 SOEM 总线在哪里结束。

After the cut, J1 shows only port 0. The open ports tell SOEM where the line ends.



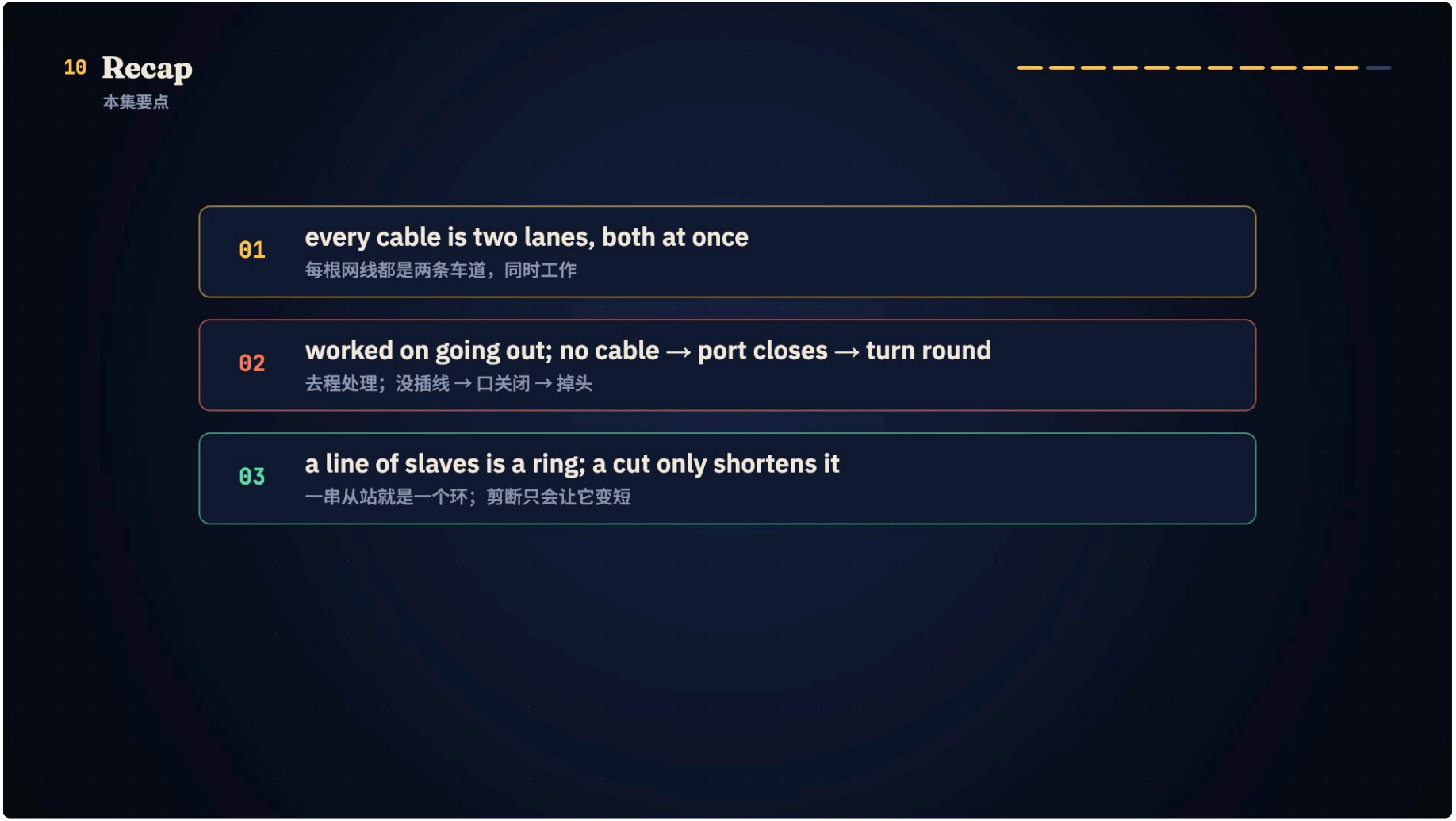
030 · 不止两个口 · More than two ports 6:49

有些从站有 3 个或 4 个口，网线就能分叉。

Some slaves have 3 or 4 ports, so the cable can branch.

还是同一条规则：每个口把帧交给下一个口。帧把每条支路挨个走完再回家：树形也还是一个环。

Same rule: each port hands the frame to the next. It walks every branch in turn, then comes home: a tree is still one ring.



031 · 本集要点 · Recap 7:10

一、每根网线都是两条车道：一对线一个方向，同时工作。

One: every cable is two lanes, one pair each way, both at once.

二、从站在去程上处理帧；没插线的口自动关闭，帧在那里掉头。

Two: the slave works on the frame on the way out. A port with no cable closes, and the frame turns around.

三、所以一串从站本身就是一个环；剪断只会让环变短。

Three: so a line of slaves is already a ring, and a cut only makes it shorter.



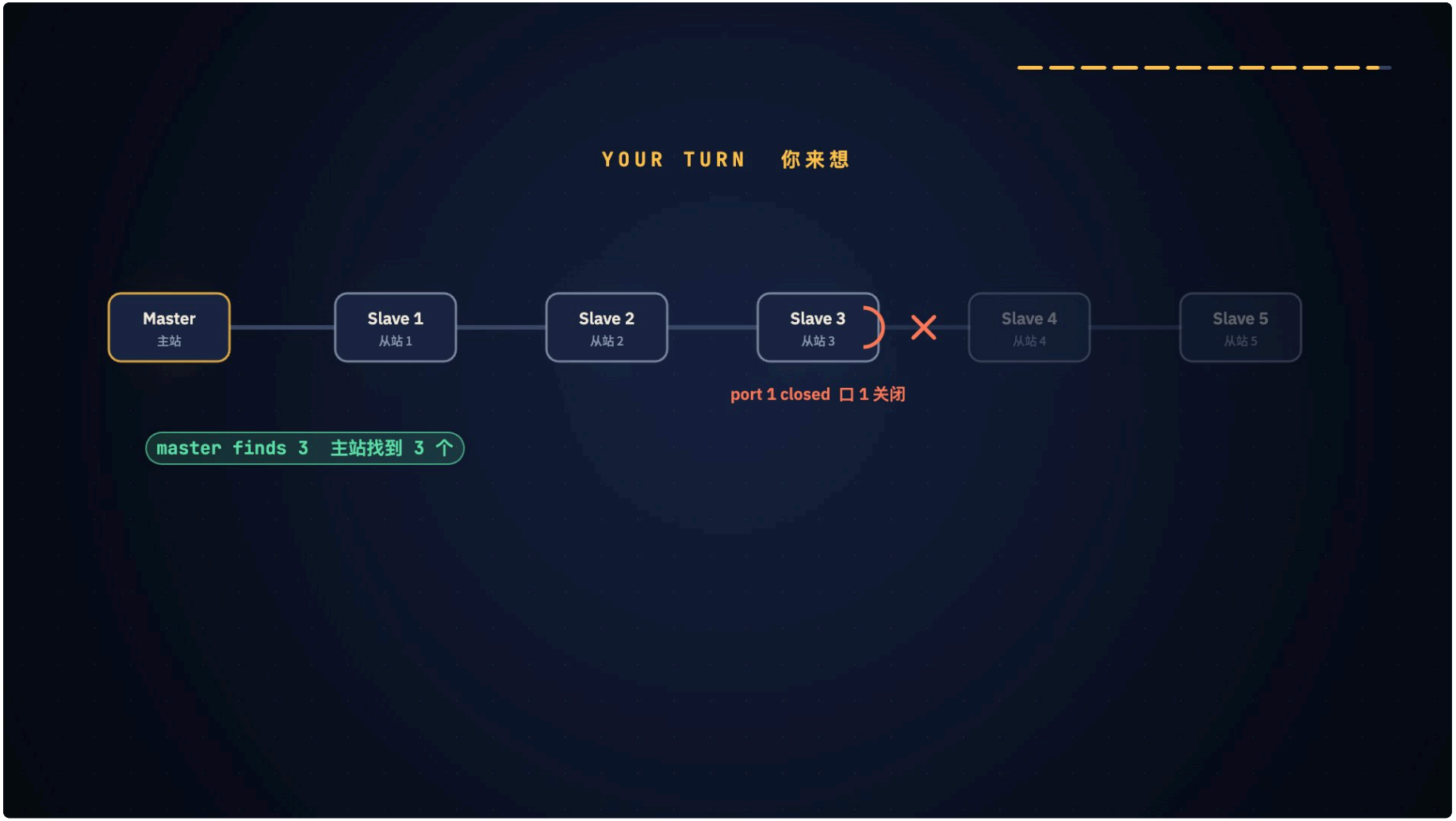
032 · 你来想 · Your turn 7:38

一条总线有 5 个从站，从站 3 后面的网线断了。

A line has 5 slaves. The cable after slave 3 is cut.

主站能找到几个从站？哪个口会关闭？

How many slaves does the master find? Which port closes?



033 · 你来想 · Your turn 7:52

3 个从站。从站 3 的口 1 关闭，帧在那里掉头。

3 slaves. Slave 3's port 1 closes, and the frame turns around there.



034 · 你来想 · Your turn 7:58

下一集：帧 1.5 微秒就回来了，可最短的帧发出去也要 5.8 微秒。

Next: the frame is back after 1.5 µs. But even the shortest frame takes 5.8 µs to send.

帧头回到了家，帧尾还没出门。第 3 集：边走边处理。

Its head is home before its tail has left. Episode 3: processing on the fly.

代码索引 · Code index

视频里出现的 3 段源码，按出场顺序。对着文件和行号打开源码，就是视频里那几行。 · The 3 source excerpts shown in the video, in order of appearance. Open the file at these lines and you see exactly what the video showed.

#	文件 · 行 · File · lines	函数 · Function	出现在 · Shown in	视频 · Video
1	soem/src/**ec_config.c** · L72		SOEM 怎么看到的 · How SOEM sees it 课件 · Slide 026	5:52
2	soem/src/**ec_config.c** · L279-292		SOEM 怎么看到的 · How SOEM sees it 课件 · Slide 028	6:26
3	soem/src/**ec_config.c** · L307		SOEM 怎么看到的 · How SOEM sees it 课件 · Slide 028	6:29

练习 · Practice

先自己答，再看答案。每个答案都写了依据：源码行，或者真实运行。 · Answer first, then open the answer. Each one says what it rests on: source lines or a real run.

第 1 题 · Q1 视频里的题 · From the video 3:10

一帧从主站出发，经过从站 1、J1、J2 再回来。它的第一个比特要多久回到主站?

A frame leaves the master, passes slave 1, J1 and J2, and comes back. How long until its first bit is back at the master?

1485 纳秒，约 1.5 微秒。去程 10 → 320 → 630 纳秒到达三个从站，在 J2 掉头，935 纳秒离开 J2，1205 离开 J1，1475 离开从站 1，1485 回到主站。

1485 ns, about 1.5 µs. Going out it reaches the three slaves at 10, 320 and 630 ns, turns round in J2, leaves J2 at 935, J1 at 1205, slave 1 at 1475, and is back at 1485.

依据 · Evidence

真实运行里每一帧都走同样的路径（data.py 逐帧检查）。延迟是模拟的：网线每米 5 纳秒；芯片去程 305 纳秒、回程 265 纳秒，取自 ET1100 数据手册的平均值。 / In the real run every frame takes the same path (data.py checks each one). The delays are simulated: 5 ns per metre of cable; 305 ns through the chip going out and 265 ns coming back, the ET1100 datasheet averages.

`episodes/ep02-one-cable-ring/data/runs.sh`

第 2 题 · Q2 视频里的题 · From the video 按代码推演 · Worked out from the code 1:36

一帧在回程上经过 J1。J1 的处理单元会再读写它一次吗?

A frame passes J1 on its way back. Does J1's processing unit read or write it again?

不会。从口 1 进来的帧不经过处理单元，直接从口 0 转发出去。每个从站只在去程上处理一次。

No. A frame coming in at port 1 skips the processing unit and is forwarded straight out of port 0. Each slave works on a frame once, on the way out.

依据 · Evidence

ESC 数据手册（第 I 部分，3.2 节，表 10）：两个口的芯片，处理顺序是 0 → 处理单元 → 1，以及 1 → 0。 / ESC datasheet (Section I, 3.2, Table 10): for two ports the order is 0 → processing unit → 1, and 1 → 0.

第 3 题 · Q3 视频里的题 · From the video 5:02

把 J1 和 J2 之间的网线剪断。主站能找到几个从站？帧在哪里掉头？一趟要多久?

Cut the cable between J1 and J2. How many slaves does the master find? Where does the frame turn round? How long is the trip?

2 个从站。J1 的口 1 失去链路自动关闭，帧在 J1 掉头。一趟 905 纳秒（完整时是 1485 纳秒）。

2 slaves. J1's port 1 loses its link and closes by itself, so the frame turns round in J1. The trip takes 905 ns (1485 ns intact).

依据 · Evidence

真实运行：VWIRE_BREAK=2 时 SOEM 的扫描输出是 2 个从站，ports 为 11 / 10。延迟同样是模拟值。 / Real run: with VWIRE_BREAK=2, SOEM's scan reports 2 slaves with ports 11 / 10. The delays are simulated, as before.

`episodes/ep02-one-cable-ring/data/runs.sh`

第 4 题 · Q4 视频里的题 · From the video 5:45

SOEM 从 J2 读回寄存器 0x0110 = 0x5611。口 1 的两位是什么？SOEM 会认为口 1 打开吗?

SOEM reads back register 0x0110 = 0x5611 from J2. What are port 1's two bits? Does SOEM count port 1 as open?

第 11、10 位是 01：关闭。SOEM 检查 (topology & 0x0c00) == 0x0800，不成立，所以口 1 不记入 activeports，扫描输出显示 10。

Bits 11 and 10 are 01: closed. SOEM tests (topology & 0x0c00) == 0x0800, which fails, so port 1 is left out of activeports and the scan shows 10.

依据 · Evidence

0x5611 的第 11:10 位：0x5611 >> 10 = 0x15，& 3 = 1，即 01。10 才表示打开且已通信。 / Bits 11:10 of 0x5611: 0x5611 >> 10 = 0x15, and & 3 = 1, that is 01. Only 10 means open and communicating.

`soem/src/ec_config.c:279` `soem/src/ec_config.c:288` `soem/src/ec_config.c:307`

`episodes/ep02-one-cable-ring/data/runs.sh`

第 5 题 · Q5 视频里的题 · From the video 按代码推演 · Worked out from the code 7:37

一条总线有 5 个从站，从站 3 后面的网线断了。主站能找到几个从站？哪个口会关闭?

A line has 5 slaves and the cable after slave 3 is cut. How many slaves does the master find? Which port closes?

3 个从站。从站 3 的口 1 关闭，帧在那里掉头；从站 4、5 收不到帧。

3 slaves. Slave 3's port 1 closes and the frame turns round there; slaves 4 and 5 get no frames.

依据 · Evidence

和剪断 J1 后面网线是同一条规则：没有链路的口自动关闭，帧交给下一个口（两个口的芯片里就是口 0）。 / The same rule as cutting after J1: a port with no link closes by itself, and the frame goes to the next port (port 0 in a two-port chip).